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Why do recycling interventions fail? A systematic literature review

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ABSTRACT

Global municipal solid waste generation is increasing from 2.1 billion tons in 2023 to 3.8 billion tons by 2050, making effective household interventions critical for sustainable waste management. However, household recycling rates remain disappointingly low despite decades of intervention efforts. This systematic literature review shifts focus from individual recycling behavior to recycling interventions themselves, examining why interventions fail rather than why people do not recycle. Following PRISMA guidelines, we searched Scopus and Web of Science, ultimately including 64 studies from 31 countries representing over 90,000 participants. We analyzed intervention types, effectiveness measurement approaches, and barriers using the macro-meso-micro framework. Results reveal a critical disconnect between intervention design and effectiveness: 71% of interventions rely on communication campaigns despite persistent evidence of the knowledge-behavior gap, while only 12.5% achieved high effectiveness (>50% improvement). Concerning patterns emerged, with 19% of studies showing declining effects over time, indicating fundamental sustainability challenges. Barrier analysis identified 150 constraints, with meso and micro-level barriers dominating. Geographic concentration in developed economies raises critical questions about applicability to developing regions where waste management challenges are most acute. These findings demonstrate that effective recycling interventions require multi-level approaches simultaneously addressing infrastructure, implementation quality and behavioral dimensions.

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1 Introduction

Global municipal solid waste generation is projected to increase from 2.1 billion tons in 2023 to 3.8 billion tons by 2050 (United Nations Environment Programme & International Solid Waste Association, 2024). This exponential growth in waste generation, coupled with shrinking landfill capacity and increasing difficulty in securing new disposal sites, has elevated recycling from an environmental preference to an operational necessity for sustainable urban development (Awino & Apitz, 2024). While the waste hierarchy framework, adopted internationally through initiatives like the EU Waste Framework Directive and UN Environment Program guidelines, positions recycling as a critical strategy for resource recovery, household recycling rates remain disappointingly low across most regions despite decades of intervention efforts (Cox et al., 2010; Sewak et al., 2021).

This disconnection between recycling's recognized importance and actual household participation represents a persistent challenge in environmental management. Previous research has extensively documented individual-level factors influencing recycling behavior (Vining & Ebreo, 1990), while more recent work identifies complex interactions between attitudes, social norms and contextual factors (Knickmeyer, 2020; Macklin et al., 2023). Passafaro and Livi (2017) demonstrated that perceived and actual recycling skills often diverge, suggesting that competence barriers interact with motivational factors in complex ways. However, this focus on understanding why individuals don't recycle has overshadowed a more fundamental question: why do recycling interventions themselves fail to achieve their intended outcomes?

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This distinction is critical because intervention failure may result from implementation barriers, design flaws, or contextual factors that are fundamentally different from individual behavioral barriers (Moeini et al., 2023).

Despite substantial theoretical advances in behavioral science, including sophisticated nudging approaches grounded in the theory of planned behavior (TPB) (Ajzen, 1991; Thaler & Sunstein, 2008) and evidence from meta-analyzes demonstrating potential effectiveness of well-designed interventions (Nisa et al., 2019; Varotto & Spagnolli, 2017), practical recycling interventions remain surprisingly traditional. Communication campaigns continue to dominate intervention strategies even as evidence mounts regarding the knowledge-behavior gap; the phenomenon where increased awareness fails to translate into behavioral change (Heidbreder et al., 2019; Parajuly et al., 2020). This persistence of information-based approaches suggests either a significant knowledge-practice gap in intervention design or systematic constraints that prevent implementation of more sophisticated behavioral strategies.

Furthermore, existing systematic reviews have primarily examined factors influencing recycling behavior rather than systematically analyzing intervention effectiveness and implementation barriers. While Varotto and Spagnolli (2017) conducted a meta-analysis finding varying effectiveness depending on contextual factors, and Sewak et al. (2021) noted disappointingly low recycling rates despite interventions, they provide limited analysis of why certain interventions fail or what prevents successful interventions from being sustained over time. Cox et al. (2010) examined policy measures but focused on identifying strategies rather than analyzing barriers to implementation, while Nisa et al. (2019) confirmed that interventions can produce behavioral changes but acknowledged persistent questions about their long-term sustainability. This gap in understanding intervention-specific barriers, as distinct from behavioral barriers, limits our ability to design more effective recycling programs.

This systematic review addresses these critical gaps by shifting focus from individual recycling behavior to recycling interventions themselves. Rather than asking 'Why don't people recycle?' We examine 'Why do recycling interventions fail?' To answer this, we leverage the vast body of existing empirical literature. While these primary studies were predominantly designed to measure individual or communal behavioral outcomes, their implementation reports, methodological designs and documented field challenges provide a rich, untapped repository of data regarding intervention mechanics. By systematically extracting this embedded information, we can evaluate implementation barriers, effectiveness measurement approaches and the multi-level constraints that prevent interventions from achieving intended outcomes. We employ the macro-meso-micro barrier framework (Caldera et al., 2022) to systematically categorize implementation challenges across structural, institutional and individual levels, providing a comprehensive understanding of intervention failure modes. By following this procedure, we aim to provide actionable insights for intervention designers and policymakers seeking to improve program outcomes rather than simply understanding individual behavior.

2 Literature review

2.1 Global waste management framework and the waste hierarchy

Waste generation and its management practices vary across both regions and income groups. Municipal solid waste management practices range from recycling and waste-to-energy solutions to more basic methods like sanitary landfilling or, in many low- and middle-income countries, uncontrolled dumping and open burning. The annual global municipal waste generation volumes are expected to increase by 73% and the volume of uncontrolled waste disposal is projected to nearly double by 2050. Higher-income regions, led by North America, generate the most municipal solid waste per capita—a trend expected to continue. Meanwhile, total waste generation is projected to rise sharply in low- and upper-middle-income countries, as well as in East Asia and the Pacific, where per capita waste levels are also expected to increase significantly. These issues pose a significant challenge for human and planetary health and place immense pressure on waste management systems. United Nations Environment Programme (2024); Kaza et al. (2021); Zhang et al. (2025).

Adopted internationally through the EU Waste Framework Directive (Environment Internet, n.d.), and the UNEP Global Waste Management Outlook—which calls to 'treat all waste materials as valuable

resources' (United Nations Environment Programme & International Solid Waste Association, 2024), the waste hierarchy establishes a priority order from most to least preferable: prevention, preparing for reuse, recycling, other recovery and disposal. Prevention involves reducing waste generation before materials turn into waste, while preparing for reuse encompasses operations that enable continued product use. Recycling, positioned in the middle of the hierarchy, involves reprocessing waste into new products or materials. Other recovery methods utilize waste to replace resources in specific functions, while disposal represents the least favorable option (Waste hierarchy Internet, n.d.).

International implementation of the waste hierarchy faces significant challenges. Awino and Apitz (2024) found that 'despite variations in drivers and need across regions, nations in both the Global North and South need to improve waste hierarchy and circular economy compliance'. Their global analysis revealed that failures to comply with waste hierarchy principles result in persistently high generation, incineration, landfill and open dump rates worldwide, highlighting the critical need for effective interventions to support hierarchy implementation.

Recycling involves separating, recovering and reusing solid waste materials with residual economic value (Nathanson, 2025). Commonly recycled items—such as paper, aluminum, glass, metals, wood and plastics—act as alternatives to raw materials sourced from dwindling natural resources like fossil fuels, mineral ores and timber (The Editors of Encyclopedia Britannica, 2025). In practice many materials can be recycled several times (Wakefield, 2022).

Households play a crucial role in recycling system effectiveness, particularly through source separation—the practice of sorting waste into distinct fractions at the point of generation. When waste sorting takes place at the household level, contamination of materials is reduced, thereby increasing the amount of recoverable recyclable material. Nathanson (2025). Effective source separation requires both increased collection rates of correctly sorted recyclable materials and reduced contamination in residual waste streams (Rousta et al., 2015).

The global scale of the waste management challenge, with costs projected to reach USD 640.3 billion annually by 2050 without urgent action (United Nations Environment Programme & International Solid Waste Association, 2024), emphasizes the critical importance of household-level interventions. Behavioral interventions, particularly nudging approaches, have emerged as promising strategies for encouraging recycling compliance within this hierarchical framework.

2.2 Behavioral approaches to recycling

Nudging, as conceptualized by Thaler and Sunstein (2008), 'involves altering the physical, social, or psychological environment in which individuals make decisions, without prohibiting options or significantly altering economic incentives'. While nudging has emerged as a potentially useful tool for promoting pro-environmental behavior in waste management policies, its actual effectiveness remains contested (Entwistle, 2023).

Research on consumer recycling behavior has predominantly drawn from the theory of planned behavior (TPB) (Ajzen, 1991, 2014), which emphasizes the role of attitudes, social norms and perceived behavioral control in predicting intentions and behaviors. Nudging interventions often incorporate social norms, though their effects remain unclear (Czajkowski et al., 2019).

Behavioral interventions for waste management can be categorized using the 4Es framework: enable, engage, encourage, and exemplify (GOV.UK Internet, 2011). These strategies operate within the established waste management hierarchy of reduce, reuse, recycle and recover (the 4Rs principle), which prioritizes interventions based on their environmental impact, with reduction being the most effective, followed by reuse, recycling and recovery (Marquez & Gomez, 2022; Kazerooni Sadi et al., 2012).

Nudging interventions for recycling can take various forms, including modifications to service provision, container sizing, equipment availability and information presentation (Stoknes, 2015). Additionally, Fun Theory approaches suggest that making pro-environmental behaviors enjoyable can be an effective strategy for behavioral change (Takahashi & Kang, 2020).

Barriers to recycling represent obstacles that impede individuals' ability or willingness to engage in waste separation and recycling behaviors (Caldera et al., 2022). These barriers can be systematically categorized into three levels: macro-barriers, which encompass structural, legal, regulatory and economic

conditions; meso-barriers, which focus on local institutional and community-level issues; and micro-barriers, which describe individual-level and day-to-day practices (Caldera et al., 2022). This multilevel framework provides a comprehensive lens for understanding the complex array of factors that can inhibit recycling initiatives.

A fundamental prerequisite for effective recycling is knowledge: individuals must be able to recognize recyclable materials and understand appropriate disposal procedures (Passafaro & Livi, 2017). Vining and Ebreo (1990) identified knowledge as a key predictor of recycling behavior, and the importance of accessible information remains relevant today. Knowledge barriers can manifest as a lack of awareness about recycling programs, confusion about which materials are recyclable, or uncertainty about proper sorting procedures. These practical barriers, along with inconvenience (Roy et al., 2022), often intersect with infrastructure limitations, creating compound obstacles to recycling participation.

2.3 Previous literature reviews: Scope and limitations

Existing systematic reviews have predominantly examined factors influencing individual recycling behavior rather than the effectiveness and barriers to recycling interventions themselves. Macklin et al. (2023) conducted an interdisciplinary review revealing many and multifaceted influences on household recycling, while Zhang (2023) categorized these influences into individual attitudes, social norms and contextual factors. Similarly, Knickmeyer (2020) focused on social factors affecting household waste separation. While these reviews provide valuable insights into behavioral determinants, they offer limited systematic analysis of intervention design, implementation, or failure modes.

Several reviews have attempted to evaluate intervention effectiveness, but with significant methodological limitations. Varotto and Spagnolli (2017) conducted a meta-analysis of psychological strategies, finding varying effectiveness depending on contextual factors, but did not systematically examine why certain interventions failed. Moeini et al. (2023) investigated household interventions but highlighted the challenge of maintaining behavioral changes without analyzing the specific barriers that undermine sustainability. Sewak et al. (2021) noted that recycling rates remain disappointingly low in many regions despite interventions, yet provided limited analysis of implementation barriers that might explain these persistent failures.

Nisa et al. (2019) confirmed that well-designed interventions can produce behavioral changes but acknowledged that questions remain about long-term sustainability once interventions conclude. This pattern across reviews suggests a critical gap: while researchers can identify effective intervention components, they have not systematically examined what prevents these interventions from succeeding in real-world implementation.

Material-specific reviews have revealed context-dependent challenges but lack systematic comparison across intervention types. Heidbreder et al. (2019) demonstrated that technical solutions alone are insufficient for plastic waste reduction, while Parajuly et al. (2020) found that awareness campaigns increase e-waste knowledge but fail to change behavior. However, these reviews do not provide frameworks for understanding when and why different intervention approaches succeed or fail across contexts.

Cox et al. (2010) examined policy measures but focused on identifying strategies rather than analyzing barriers to policy implementation. Together, these gaps signal that a review focused specifically on intervention failure mechanisms is needed.

3 Critical research gaps and rationale for the current study

The synthesis of existing literature reveals three significant limitations that justify the current study's approach:

Gap 1) Intervention-centered vs. behavior-centered analysis. Existing reviews primarily ask, ‘why don't people recycle’ rather than ‘why do recycling interventions fail’. This distinction is crucial because intervention failure may result from implementation barriers, design flaws, or contextual factors that are distinct from individual behavioral barriers.

Gap 2) Limited systematic analysis of effectiveness measurement. While multiple reviews acknowledge varying intervention effectiveness, none have systematically examined how effectiveness is measured across

studies or whether measurement approaches themselves contribute to inconsistent findings about intervention success.

Gap 3) Absence of the barrier-to-intervention-success framework. Existing reviews identify barriers to recycling behavior but do not examine barriers that prevent interventions from achieving their intended effects. This represents a fundamental gap in understanding how to design more successful recycling programs.

This systematic review addresses these gaps by focusing on recycling interventions rather than recycling behavior, examining how intervention effectiveness is measured and reported, and developing a framework for understanding barriers to intervention success rather than barriers to individual recycling. This approach provides actionable insights for intervention designers and policymakers seeking to improve program outcomes rather than simply understanding individual behavior.

Crucially, while the primary aim of much of the existing literature has been to evaluate individual behavioral shifts, these studies inherently serve as detailed repositories of intervention mechanics. In the process of testing behavioral theories, these primary studies meticulously document their intervention designs, implementation procedures, measured outcomes and the practical challenges encountered in the field. Therefore, by systematically extracting and synthesizing this secondary data—repurposing the structural and operational details embedded within behavior-focused studies—the existing literature is uniquely equipped to answer fundamental questions about intervention architecture and implementation failure.

Three research questions guided the review:

RQ1 What kind of recycling interventions are conducted to promote the recycling of households?

RQ2 How is the effectiveness of the interventions measured, and what is the outcome of the interventions?

RQ3 What barriers to intervention effectiveness emerge, and how do they map onto macro-, meso-, and micro-level categories?

4 Methods

4.1 Study design

This study employed a systematic literature review methodology following the Preferred Reporting Items for Systematic Reviews and Meta-Analyzes (PRISMA) guidelines to investigate barriers in the recycling process of consumers (Page et al., 2021).

4.2 Search strategy

A comprehensive literature search was conducted on 14 February 2024, using two electronic databases: Scopus and web of science (WoS). This approach aligns with recommendations to use multiple databases to ensure comprehensive coverage (Bramer et al., 2017). The search strategy was developed around three thematic areas: recycling, nudging and barriers for change. The search terms utilized are presented in Table 1.

The full search string was constructed using Boolean operators to combine terms within and across thematic areas (Methley et al., 2014):

((‘recycling’ OR ‘circular economy’ OR ‘waste management’ OR ‘resource recovery’ OR ‘sustainable waste practice’ OR ‘eco-friendly disposal’ OR ‘material reclamation’ OR ‘waste reduction’ OR ‘reuse and repurpose’ OR ‘waste diversion’ OR ‘recyclable material*’ OR ‘waste sorting’ OR ‘eco-conscious disposal’

Table 1. Search terms by thematic area.

Thematic area	Search terms
Recycling	‘recycling’, ‘circular economy’, ‘waste management’, ‘resource recovery’, ‘sustainable waste practice’, ‘eco-friendly disposal’, ‘material reclamation’, ‘waste reduction’, ‘reuse and repurpose’, ‘waste diversion’, ‘recyclable material*’, ‘waste sorting’, ‘eco-conscious disposal’, ‘closed-loop systems’, ‘waste separation’, ‘upcycling’, ‘waste-to-energy’
Intervention	‘*intervention’, ‘*nudging’, ‘*nudge’, ‘behavior change program’, ‘social change initiative’, ‘green practice’, ‘*campaign’
Barriers for change	‘barrier*’, ‘resistance to change’, ‘obstacle*’, ‘factor*’, ‘hurdle*’, ‘bottleneck*’, ‘challenge*’

OR 'closed-loop systems' OR 'waste separation' OR 'upcycling' OR 'waste-to-energy') AND ('intervention' OR 'nudging' OR 'nudge' OR 'behavior change program' OR 'social change initiative' OR 'green practice' OR 'campaign') AND ('barrier' OR 'resistance to change' OR 'obstacle' OR 'factor' OR 'hurdle' OR 'bottleneck*' OR 'challenge*')).

Studies were selected based on a predefined set of inclusion criteria (Table 2), which were developed following guidelines for systematic reviews in environmental science (Pullin et al., 2022).

Challenges related to recycling were assessed during the full-text review phase to ensure comprehensive understanding of the barriers identified in the literature.

The initial search yielded 2026 records from Scopus and 1874 records from Web of Science. After removing duplicates, titles and abstracts were screened independently by two reviewers to determine eligibility for full-text review, following best practices for systematic reviews (Higgins et al., 2019). Disagreements were resolved through discussion until consensus was reached. Inter-rater reliability was calculated using Cohen's Kappa coefficient to assess the consistency of selection decisions between reviewers (McHugh, 2012). For the abstract screening phase, and full-text screening phase, two reviewer pairs conducted the assessment.

Following the abstract screening, full-text articles were retrieved and independently assessed for eligibility by two reviewers. Reasons for exclusion were documented during this phase. The entire study selection process was documented using a PRISMA flow diagram (Page et al., 2021) (Figure 1). After this process, there was a total of 64 records included in the review.

To evaluate the methodological quality and risk of bias of the highly heterogeneous included studies, we employed the Mixed Methods Appraisal Tool (MMAT) version 2018. The MMAT is specifically designed for systematic reviews that integrate qualitative, quantitative and mixed-methods study designs. Reviewers independently assessed each study across five methodological criteria specific to its study category. Disagreements were resolved through discussion. Studies were scored by calculating the percentage of criteria met and subsequently categorized into three quality tiers: High quality (meeting 80–100% of criteria), Moderate quality (meeting 60%) and low quality (meeting 40% or fewer).

4.3 Data analysis

Data from eligible studies were extracted using a standardized extraction form. To ensure accuracy, one reviewer extracted the data and a second reviewer verified the extracted information for completeness and correctness. Any discrepancies were resolved through discussion. The extraction process captured information relevant to the research questions, including details about recycling initiatives, motivational factors for consumers and barriers to effective recycling. This structured approach to data extraction ensured consistency across studies and reviewers (Munn et al., 2018).

Across the studies included, waste fractions were operationalized at varying levels of granularity, with the fractions and terminology summarized in Table 3. For paper, studies variously used broad labels (e.g. 'paper') or more specific terms such as newsprint/newspaper as the sole paper fraction. Cardboard was often defined broadly to encompass cardboard and paper packaging materials. Plastics ranged from generic labels (e.g. 'plastic' or 'plastic waste/packaging') to polymer-specific classifications (e.g. PE, PET, PP, HDPE). Bottles and cans were inconsistently specified: some studies reported only 'bottles' and/or 'cans' without stating the underlying material, whereas others distinguished by material type. For the

Table 2. Eligibility criteria.

Criterion	Inclusion	Exclusion
Target population	Households	Restaurants, construction industry, industrial settings, healthcare facilities, businesses, entrepreneurs, business developers, policy makers, schools
Waste types	Biowaste, paper, cardboard, plastic, metal, electronics, glass, hazardous waste, medication, household construction waste	Industrial waste, restaurant waste, commercial waste, construction waste (non-household originated)
Intervention	Studies with nudging or behavior change interventions	Modeling studies without interventions, Interventions focusing on other levels of waste hierarchy (e.g. reducing waste), Studies focusing on intention without behavioral change
Language	English	Non-English
Publication types	Original studies (journal articles, conference papers)	Books, theses, systematic reviews

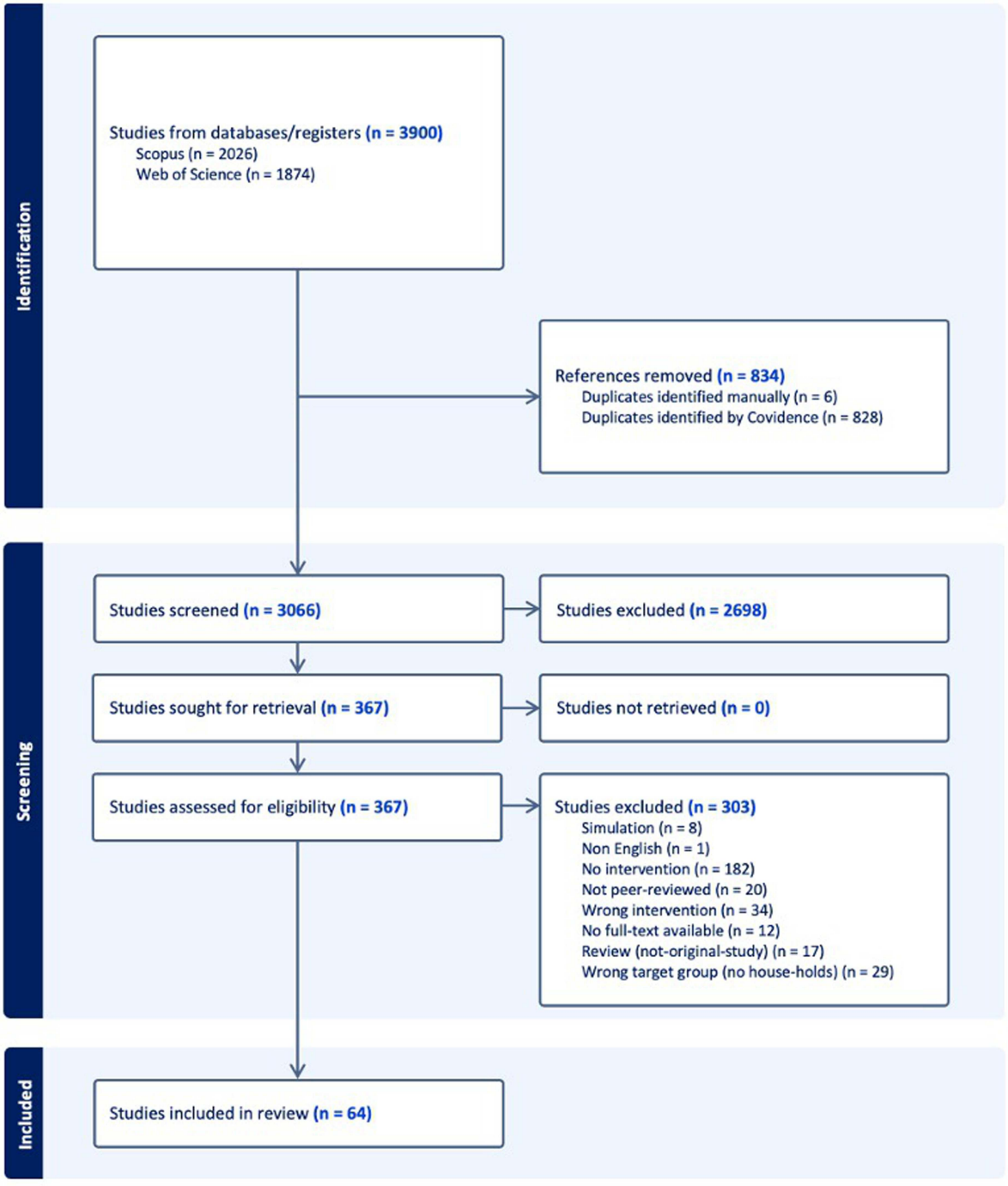


Figure 1. The flow diagram of the systematic literature review following the PRISMA procedure.

purposes of this review, bottles were coded under glass or plastic when the material was explicitly stated and cans were coded under metal, reflecting the common classification of cans as metal regardless of the specific type mentioned (e.g. tin or aluminum). Finally, while batteries were in some studies listed as standalone items and in others referenced in connection with hazardous waste, they were coded under hazardous waste in this review to maintain consistent classification.

A substantial subset of studies either did not specify fractions beyond general waste-stream terms (e.g. 'household waste', 'solid waste', 'dry waste', 'residual waste', 'mixed waste'), sometimes with only illustrative examples, or included an explicit residual/mixed category capturing the remainder not covered by separately identified fractions. The likely composition of these residual categories varied by study,

Table 3. Waste fraction terminology across included studies.

Waste fraction	Terms and representative examples
Paper	Paper/paper waste Cole et al. (2014); Cotterill et al. (2009); Chase et al. (2009); Dhokhikah et al. (2015); Del Cimmuto et al. (2014) Newsprint/newspaper Rousta et al. (2015); Mehner et al. (2020); Bernstad et al. (2013); Vining and Ebreo (1989)
Metal	Metal(s) Cole et al. (2014); Hernández et al. (1999); Xu et al. (2023); Meen-Chee and Narayanan (2006); Willman (2015) Metal packaging Rousta et al. (2015); Mehner et al. (2020) Cans Park and Lah (2015); Hotta and Aoki-Suzuki (2014); Kutralam-Muniasamy et al. (2023); Timlett and Williams (2008) Tin cans Vining and Ebreo (1989) Aluminum Vining and Ebreo (1989); Okonta and Mohlalifi (2020)
Organic waste	Organic waste/organic matter/organics Lakhan (2016); Geislar (2017); Read et al. (2009); Del Cimmuto et al. (2014); Manomaivibool et al. (2018); Pongpunpurt et al. (2022) Food waste Rousta et al. (2015); Löytty et al. (2023); Bian et al. (2022); Roe et al. (2022); Linder et al. (2018); Sintov et al. (2019) Wet waste Xu et al. (2018) Kitchen waste Liang et al. (2024) Compost/compostable Chase et al. (2009); Nguyen (2022)
Glass	Glass Rousta et al. (2015); Pegels et al. (2022); Vining and Ebreo (1989); Meen-Chee and Narayanan (2006); Timlett and Williams (2008) Glass packaging Rousta et al. (2015); Mehner et al. (2020) Glass bottles Kutralam-Muniasamy et al. (2023); Cotterill et al. (2009)
Cardboard	Cardboard Cole et al. (2014); Kutralam-Muniasamy et al. (2023); Flygansvær et al. (2021); Vining and Ebreo (1989); Meen-Chee and Narayanan (2006); Kattoua et al. (2019) Card/thin card Cotterill et al. (2009); Timlett and Williams (2008) Paper packaging Rousta et al. (2015); Mehner et al. (2020); Lakhan (2016)
Hazardous waste	Hazardous waste Díaz et al. (2012); Xu et al. (2018); Liang et al. (2024); Pongpunpurt et al. (2022), often paired with mentions of specific types of waste such as aerosol cans, fluorescent lamp and batteries
Plastic	Plastic(s) Bernstad et al. (2013); Park and Lah (2015); Lakhan (2016); Pegels et al. (2022); Meen-Chee and Narayanan (2006); De Fano et al. (2022); Del Cimmuto et al. (2014); Okonta and Mohlalifi (2020) Plastic items Cole et al. (2014) Plastic packaging Rousta et al. (2015); Mehner et al. (2020); Flygansvær et al. (2021) Specific polymers e.g. PE, PET, PP, HDPE Kutralam-Muniasamy et al. (2023); Pongpunpurt et al. (2022) Plastic bottles Xu et al. (2018); Dutta-Powell et al. (2023); Cotterill et al. (2009); Timlett and Williams (2008) PET bottles Hotta and Aoki-Suzuki (2014)
E-waste	E-waste Kutralam-Muniasamy et al. (2023); Sabbir et al. (2023); Otto et al. (2018); Nowakowski et al. (2021) Electronics Rousta et al. (2015) Mobile phones Welfens et al. (2016)
Textiles	Textiles Park and Lah (2015); Roe et al. (2022); Cotterill et al. (2009); Timlett and Williams (2008) Clothes/old clothes Dutta-Powell et al. (2023); Pongpunpurt et al. (2022)
General waste stream	Household waste Li (2003); Remr (2018); Linder et al. (2018); Tong et al. (2018) Solid waste Liu et al. (2023); Nepal et al. (2023); Omran et al. (2009); Read (1999) Residual waste/residues Bernstad et al. (2013); Díaz et al. (2012); Bian et al. (2022); Flygansvær et al. (2021); Del Cimmuto et al. (2014) Mixed waste Cole et al. (2014) Dry waste Xu et al. (2018); Dutta-Powell et al. (2023); Xu et al. (2023); Ling et al. (2023) Other waste Read et al. (2009); Liang et al. (2024); Okonta and Mohlalifi (2020); Pongpunpurt et al. (2022) Non-organic recyclables/non-compostable Schmidt et al. (2022); Nguyen (2022) Recyclable waste/recyclables/dry recyclables/other recyclable items Park and Lah (2015); Lakhan (2016); Xu et al. (2018); Read et al. (2009); Nguyen (2022); Pongpunpurt et al. (2022); Willman (2015) Combustible waste/waste to incinerate Mehner et al. (2020); Hernández et al. (1999) Incombustibles Hernández et al. (1999) Rubbish/garbage Read (1999); Willman (2015)

reflecting differences in local sorting policies, collection infrastructure and the degree of source separation implemented in the study context. Accordingly, Table 3 does not exhaustively list every material mentioned in the included studies but focuses on fractions that could be unambiguously classified under the general waste-fraction categories used in this review.

A narrative synthesis approach was employed to analyze and synthesize the findings from the included studies (Popay et al., 2006). The synthesis was organized around the three research questions, with particular emphasis on identifying patterns in barriers that prevent effective recycling initiatives and understanding factors that motivate or inhibit consumer recycling behavior. Furthermore, the synthesis framework was designed to extrapolate connecting themes across different intervention contexts, while explicitly mapping the unique and sometimes conflicting disciplinary perspectives (e.g. engineering, behavioral economics, environmental psychology) embedded within the literature.

4.3.1 Intervention analysis (RQ1)

Interventions were initially coded into eight primary categories based on their dominant mechanism: Communication campaign, Infrastructure development, economic incentives, system-level reformation,

interactive/engagement activities, performance-based messaging, social norm interventions and reverse logistics/product returns. Multiple coding was allowed when studies employed multi-component approaches. The coded interventions were then mapped onto the 4Es behavioral change framework (GOV.UK Internet, 2011) to identify patterns in intervention design approaches. *Enable*: infrastructure modifications, accessibility improvements; *Engage*: educational campaigns, awareness initiatives; *Encourage*: economic incentives, performance feedback; *Exemplify*: social norm interventions, comparative messaging.

4.3.2 Effectiveness assessment and synthesis (RQ2)

Intervention effectiveness was assessed through systematic extraction of quantitative and qualitative outcomes reported in each study. To structure the analysis, studies were categorized by measurement approach: self-reported (surveys/questionnaires), objective waste measurements (compositional analysis, tonnage), or mixed methods. For quantitative studies, key metrics such as percentage improvements in recycling rates, participation and waste diversion were extracted where available. For qualitative studies, reported behavioral changes and participant responses were synthesized narratively. Based on the reported outcomes and statistical significance, interventions were classified into four effectiveness levels: *high effectiveness* (>50% improvement or strong qualitative evidence), *moderate effectiveness* (15–50% improvement), *low effectiveness* (<15% improvement), or mixed results. The classification enabled a systematic synthesis of findings to identify which intervention types were most consistently associated with successful outcomes.

4.3.3 Barrier classification and analysis (RQ3)

Barriers to intervention effectiveness were systematically extracted and analyzed using the three-level framework proposed by Caldera et al. (2022). Barriers were classified as: *macro-level barriers* encompassing structural, legal, regulatory and economic policy conditions; *meso-level barriers* focusing on local institutional, infrastructure and community-level implementation issues; and *micro-level barriers* describing individual-level and household day-to-day practices. When barriers spanned multiple levels, they were assigned to the level where the primary constraint operated. The frequency of barrier mentions across studies was tabulated to identify the most commonly reported implementation challenges, with particular attention to how barrier patterns related to intervention types and effectiveness outcomes.

5 Results

The studies within the scope of this review were conducted across 31 countries (Figure 2). Asia accounts for the largest share with 30 studies (47%), followed by Europe ($n = 20$, 31%) and North America ($n = 11$, 17%). South America, Africa and Australia/Oceania were underrepresented with 3, 1 and 1 studies, respectively. At the national level, China dominated with 12 (19%) studies, followed by the United States ($n = 8$, 13%), the United Kingdom ($n = 6$, 9%), Sweden ($n = 5$, 8%) and Thailand ($n = 4$, 6%). The remaining countries contributed 1–2 studies each. One study was a cross-cultural comparison conducted in Japan, Canada and the United States. As the intervention was implemented separately in all three countries, each was marked individually in Figure 2. Consequently, the total number of map entries is 66, although the number of distinct studies is 64. This geographic concentration suggests a potential limitation in generalizability, as recycling intervention research appears heavily focused on developed economies in Asia and Europe. The underrepresentation of studies from developing regions may limit understanding of intervention effectiveness across diverse socioeconomic and infrastructure contexts.

The studies examined a diverse range of recyclable materials with plastic being the most frequently addressed material (28 studies, 44%), followed by paper ($n = 27$, 42%) and organic waste ($n = 26$, 41%). Traditional recyclables, like glass ($n = 20$, 31%), metal ($n = 19$, 30%) and cardboard ($n = 15$, 23%), were also well-represented, as well as any type of bottle ($n = 17$, 27%). Specialized waste streams received less attention, hazardous waste including batteries ($n = 13$, 20%) and electronic waste ($n = 6$, 9%). An additional category of 'other' materials encompassed diverse waste streams that fell outside the main recyclable categories, appearing across 15 (23%) study mentions. This heterogeneous grouping included textiles and clothing (the most frequently cited within this category), automotive materials (motor oil, rubber),

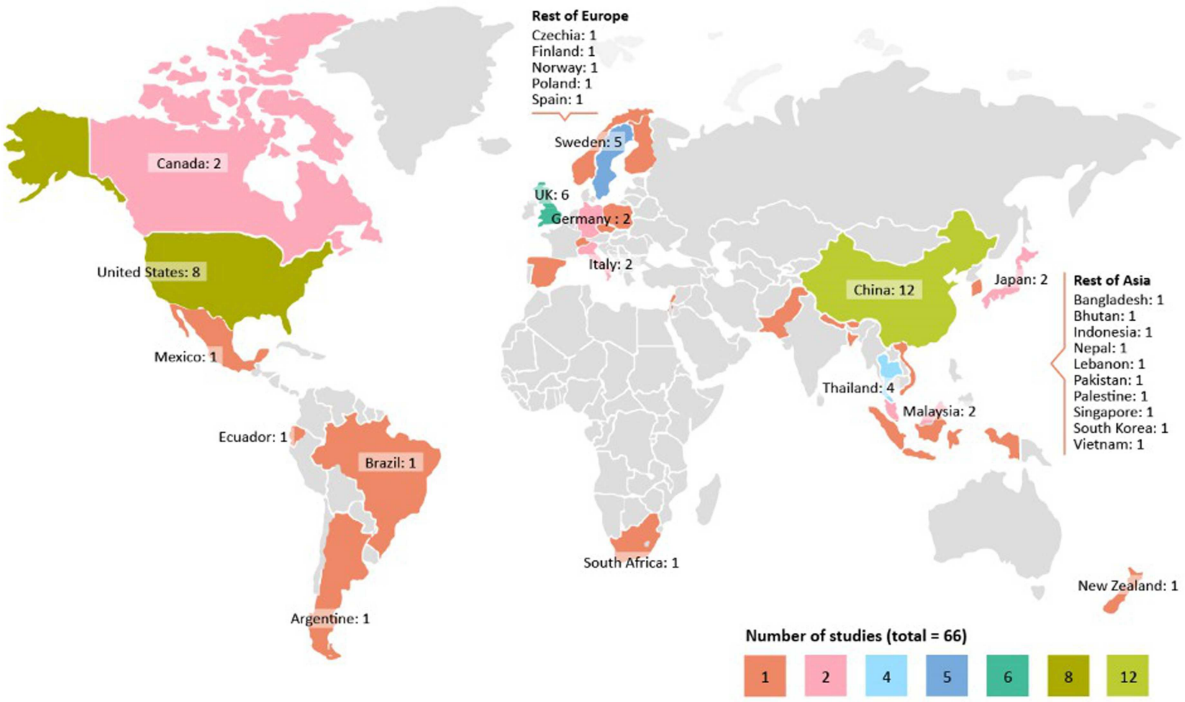


Figure 2. Geographical distribution of studies by country.

household chemicals (cooking oil, spray cans), specialized packaging (tetrapak containers) and context-specific waste types such as infectious waste and office materials.

When examining studies that focus exclusively on a single material type, organic/food waste dominated, with seven dedicated studies, reflecting growing policy attention to food waste reduction. Electronic waste and plastic waste each received focused attention in four studies, indicating particular research interest in these challenging waste streams due to their environmental impact and recycling complexity. Other material types do not appear as the sole focus of any study within the dataset, indicating that they were either examined in combination with other materials or not highlighted as distinct categories.

The methodological quality of the included studies was generally robust, indicating an acceptable overall risk of bias. Based on the MMAT appraisal, over half of the studies ($n = 34$, 53%) were classified as high quality, demonstrating strong methodological validity and reliable measurement approaches within their respective study designs. Twenty-one studies (33%) were appraised as moderate quality, typically due to unclear sampling strategies, incomplete outcome data, or lack of clarity regarding confounders. Nine studies (14%) were classified as low quality, primarily stemming from significant risks of non-response bias and poorly defined data collection methods. The predominance of high and moderate-quality evidence supports the overall credibility of the synthesized findings regarding intervention effectiveness and implementation barriers. Full quality appraisal scores for each study are available in Appendix B.

5.1 RQ1: The types of interventions

Communication campaigns dominated the intervention landscape (Table 4), appearing in 71% of studies with varying delivery mechanisms ranging from interpersonal door-stepping initiatives (Bernstad et al., 2013) to digital media campaigns (Chen et al., 2022). This heavy reliance on information-based approaches aligns with traditional assumptions that knowledge gaps drive recycling behavior, though this contradicts emerging evidence about the knowledge-behavior gap identified in the literature.

Infrastructure development interventions (38% of studies) addressed practical barriers through system modifications, from simple bin placement changes (Lakhan, 2016) to comprehensive collection system

Table 4. Intervention type distribution and characteristics.

Intervention category	Frequency	Key characteristics	Example studies
Communication campaign	47 studies (71%)	Includes education, awareness and information provision	Bowman et al. (1998); Chen et al. (2022)
Infrastructure development	25 studies (38%)	Bin placement, collection system modifications, facility improvements	Rousta et al. (2015); Bian et al. (2022)
Economic incentives	15 studies (23%)	Financial rewards, penalties, pay-as-you-throw systems	Park and Lah (2015); Li (2003)
System-level reformation	8 studies (12%)	Policy changes, regulatory modifications and program restructuring	Cole et al. (2014); Liu et al. (2023)
Interactive/engagement activities	6 studies (9%)	Participatory approaches, community engagement and gamification	Löytty et al. (2023); Roe et al. (2022)
Performance-based messaging	4 studies (6%)	Feedback on behavior, comparative information and progress tracking	Dutta-Powell et al. (2023); Xu et al. (2023)
Social norm interventions	2 studies (3%)	Normative messaging, social comparison	Bowman et al. (1998); Siepelmeyer and Otterbring (2022)
Reverse logistics/product return	3 studies (5%)	Take-back programs, manufacturer responsibility	Sabbir et al. (2023); Welfens et al. (2016)

restructuring (Geislar, 2017). These interventions directly target the 'enable' component of the 4Es framework, suggesting growing recognition that structural barriers require practical solutions.

Economic incentives appeared in 23% of studies, utilizing various mechanisms from cash-equivalent point systems (Xu et al., 2018) to waste-for-goods exchange programs (Kutralam-Muniasamy et al., 2023). The prevalence of economic approaches reflects attempts to address motivational barriers, though implementation varied significantly in scope and sustainability.

Notably underrepresented were sophisticated behavioral interventions. Social norm interventions appeared in only 3% ($n = 2$) of studies despite substantial theoretical support, while performance-based messaging ($n = 4$, 6%) and interactive engagement ($n = 6$, 9%) remained limited. This pattern suggests a significant gap between theoretical understanding of behavioral change mechanisms and practical intervention design.

System-level reformation interventions ($n = 8$, 12%) typically involved policy or regulatory changes, from mandatory separation programs (Liu et al., 2023) to comprehensive waste management restructuring (Hernández et al., 1999). These interventions address macro-level barriers but appear less frequently than micro-level approaches.

Integration patterns revealed that 65% of studies combined multiple intervention types, most commonly pairing communication campaigns with infrastructure development (18 studies, 28%). This multi-component approach suggests implicit recognition that a single-intervention strategy may be insufficient for complex behavioral change challenges.

5.2 RQ2: The effectiveness of the interventions

The reviewed studies demonstrated substantial heterogeneity in both measurement approaches and effectiveness outcomes (Table 5). Self-reported measures through questionnaires and surveys dominated the methodological landscape (54% of studies), while objective waste measurements comprised 29% of studies. Mixed-methods approaches, combining both self-reported and objective measurements, were employed in 17% of studies, providing the most comprehensive assessment of intervention impacts.

Eight studies demonstrated high effectiveness with improvements exceeding 50% in target behaviors. These high-impact interventions shared several characteristics: they typically combined multiple intervention components, addressed structural barriers alongside behavioral ones and employed objective measurement approaches. Park and Lah (2015) documented perhaps the most dramatic improvement, with recycling rates increasing from 6.3% to 47% following economic incentive implementation. Similarly, Liu et al. (2023) found participation rates rose from 26.4% to 96.43% over seven years through combined infrastructure development and system-level reformation. Notably, six of the eight high-effectiveness studies included infrastructure modifications as a core component, suggesting that addressing practical barriers may be essential for achieving substantial behavioral change.

The largest group of studies ($n = 23$, 36%) achieved moderate effectiveness with improvements between 15–50%. These interventions typically combined communication campaigns with either infrastructure

Table 5. Intervention effectiveness by category and measurement approach.

Effectiveness level	N	Measurement approaches	Key quantitative results	Example studies
High effectiveness (>50% improvement)	8	Mixed methods (5), Objective measures (2), Self-report (1)	50–96% improvements in target behaviors; Park and Lah (2015): 6.3% to 47% recycling rate; Liu et al. (2023): 26.4% to 96.43% participation	Park and Lah (2015); Liu et al. (2023); Roe et al. (2022); Otto et al. (2018)
Moderate effectiveness (15–50% improvement)	23	Self-report (15), Mixed methods (5), Objective measures (3)	15–50% improvements; typical range 20–40%; Cole et al. (2014): +91–111t/month recycling; Flygansvær et al. (2021): +17% food waste sorting	Chen et al. (2022); Cole et al. (2014); Geislar (2017); Flygansvær et al. (2021)
Low effectiveness (<15% improvement)	8	Self-report (6), Mixed methods (2)	<15% improvements or nonsignificant results; Read et al. (2009): ~2% waste reduction; Nepal et al. (2023): 7.6–11.4% increase	Bian et al. (2022); Read et al. (2009); Nepal et al. (2023)
Mixed/declining results	12	Mixed methods (7), Self-report (4), Objective measures (1)	Initial improvements followed by decline; sustainability issues; Cotterill et al. (2009): effects declined after 3 months; Ling et al. (2023): 40% to 20% effectiveness over time	Xu et al. (2023); Cotterill et al. (2009); Ling et al. (2023)
Not quantified	13	Self-report (11), Mixed methods (2)	Qualitative improvements in knowledge, attitudes, or behavior reported without specific metrics	Bortoleto and Hanaki (2007); De Gisi et al. (2017); Karout and Altuwaijri (2012)

development or economic incentives. Cole et al. (2014) demonstrated the potential of system-level changes, achieving increases of 91–111 tons per month in recycling through comprehensive program restructuring. However, effectiveness within this range was strongly influenced by contextual factors including baseline recycling rates, demographic characteristics, and existing waste management infrastructure. Studies in regions with minimal existing recycling infrastructure often showed larger percentage improvements than those in areas with established systems.

A concerning pattern emerged among 12 studies showing mixed or declining results over time. Multiple interventions demonstrated initial promise followed by diminishing impacts as intervention periods extended. Cotterill et al. (2009) observed significant participation declines three months post-intervention, while Ling et al. (2023) documented normative feedback effectiveness decreasing from 40% to 20% over the study period. This sustainability challenge appeared particularly pronounced among communication-only interventions, while infrastructure-based changes showed more durable effects. Xu et al. (2023) identified a specific mechanism underlying sustainability problems: economic incentives increased recycling behavior but simultaneously reduced intrinsic environmental motivation, resulting in a 36% reduction in program effectiveness through motivational crowding-out.

Studies employing objective measurements or mixed methods more frequently reported high effectiveness outcomes compared to self-report-only studies. This pattern may reflect both measurement precision and intervention design sophistication, as studies investing in objective measurement often implemented more comprehensive interventions. However, the knowledge-behavior gap persisted across measurement approaches: interventions frequently increased awareness and understanding without producing corresponding behavioral changes, particularly among communication-focused interventions.

5.3 RQ3: Barriers preventing recycling

The systematic analysis revealed 150 barrier mentions across the 64 studies, distributed across macro ($n = 19$), meso ($n = 65$) and micro ($n = 66$) levels (Table 6). This distribution indicates that implementation challenges predominantly occur at local institutional and individual levels rather than through high-level policy constraints, suggesting that intervention success depends heavily on addressing practical and personal barriers alongside systemic ones.

Meso-level barriers represented the largest category overall, with practical implementation issues being the most frequently reported barrier across all studies ($n = 28$, 44%). These implementation challenges encompassed diverse operational failures including service delivery problems, staff behavior issues and system coordination difficulties. Hernández et al. (1999) identified infrequent collection schedules and local collector traditions as primary obstacles, while Cotterill et al. (2009) documented how staff rudeness and missed collections undermined intervention effectiveness even when infrastructure was adequate.

Barriers related to infrastructure and facilities ($n = 24$, 38%) represented the second most common constraint, highlighting how physical systems limitations can prevent even well-designed interventions

Table 6. Barriers to recycling intervention effectiveness by multi-level framework.

Barrier category	Level	Count	Primary characteristics	Representative examples
System/policy issues	Macro	12	National regulations, policy frameworks and institutional mandates	Lack of national regulations Hotta and Aoki-Suzuki (2014); inadequate policy support Pegels et al. (2022); regulatory gaps Nepal et al. (2023)
Economic policy factors	Macro	7	Structural economic conditions, pricing policies and funding mechanisms	Cost of services Bryce et al. (1997); lack of economic incentives Park and Lah (2015); funding constraints Otto et al. (2018)
Practical implementation issues	Meso	28	Service delivery, local operations, staff performance	Infrequent collection Hernández et al. (1999); rude staff Cotterill et al. (2009); missed collections Dutta-Powell et al. (2023)
Infrastructure/facilities	Meso	24	Local infrastructure, collection systems and facility availability	Missing bins Geislar (2017); inadequate collection points Lakhan (2016); facility accessibility Omran et al. (2009)
Community economic factors	Meso	4	Local economic conditions, service costs and community resources	Unwillingness to pay for services Bryce et al. (1997); local economic constraints Li et al. (2021)
Social/cultural factors	Meso	9	Community norms, local traditions, social practices	Established collection traditions Li (2003); community resistance Nepal et al. (2023); local cultural practices Hernández et al. (1999)
Knowledge/awareness	Micro	18	Individual understanding, information access and education	Language barriers Chase et al. (2009); lack of sorting knowledge Dhokhikah et al. (2015); unclear instructions Bowman et al. (1998)
Convenience/time	Micro	21	Individual effort, time constraints and personal convenience	Distance to collection points Mehner et al. (2020); time requirements Vining and Ebreo (1989); inconvenient scheduling Read (1999)
Space constraints	Micro	14	Household storage, physical limitations and space management	Limited home storage Meen-Chee and Narayanan (2006); apartment size constraints Mehner et al. (2020); multiple bin requirements Dhokhikah et al. (2015)
Attitudes/motivation	Micro	13	Individual attitudes, personal motivation and behavioral resistance	Lack of interest Omran et al. (2009); reluctance to change Remr (2018); low environmental concern Díaz et al. (2012)

from succeeding. These barriers ranged from basic bin availability (Geislar, 2017) to complex access issues at collection points (Lakhan, 2016). The technology adoption resistance noted by Löytty et al. (2023), including 'big brother aversion' toward IoT composting systems, illustrates how infrastructure barriers extend beyond physical constraints to include technological acceptance challenges.

Micro-level barriers clustered around three primary themes affecting individual participation. Convenience and time constraints ($n = 21$, 33%) emerged as substantial obstacles, with participants reporting difficulties ranging from 500-meter walking distances to collection points (Mehner et al., 2020) to storage challenges in small apartments requiring multiple waste fraction containers. These convenience barriers often interacted with space constraints ($n = 14$, 22%) to create compound obstacles particularly challenging for urban households.

Knowledge and awareness barriers ($n = 18$, 28%) persisted despite the communication campaign's prevalence, suggesting that information provision alone may be insufficient. These barriers manifested as language difficulties (Chase et al., 2009), unclear sorting procedures (Bowman et al., 1998) and persistent confusion about recyclable materials (Dhokhikah et al., 2015). The knowledge-behavior gap identified in effectiveness analysis appears linked to these micro-level information processing challenges.

Attitudes and motivation barriers ($n = 13$, 20%) revealed deeper psychological obstacles, including environmental indifference (Omran et al., 2009), behavioral resistance (Remr, 2018) and what participants characterized as 'I don't care' responses (Díaz et al., 2012). Entwistle (Entwistle, 2023) identifies three types of fatalism—passive, protective and pathological—suggesting that motivational barriers may require differentiated intervention approaches rather than universal behavior-change strategies.

Macro-level barriers, while less frequently reported, represented fundamental constraints that can undermine local interventions. System and policy issues ($n = 12$, 19%) included regulatory gaps (Nepal et al., 2023), lack of national frameworks (Hotta & Aoki-Suzuki, 2014) and institutional mandate limitations. Economic policy factors ($n = 7$, 11%) encompassed structural pricing issues and funding constraints that create systematic obstacles to intervention sustainability.

The barrier analysis revealed important interactions between levels that may explain intervention failures. Economic factors appeared at both macro and meso levels, suggesting that financial constraints operate through multiple pathways—from policy-level funding decisions to community-level service costs to individual household economic pressures. Similarly, social and cultural factors spanned meso and micro

levels, indicating that community norms influence individual behavior while individual attitudes aggregate to shape community practices.

This multi-level barrier framework suggests that effective recycling interventions must address constraints operating simultaneously across macro, meso and micro levels rather than focusing on single-level solutions. The predominance of meso and micro barriers indicates that even well-designed policies require careful attention to implementation details and individual participation constraints to achieve intended outcomes.

6 Discussion

This systematic review reveals a fundamental disconnect between theoretical understanding of behavioral change mechanisms and practical intervention implementation. Despite decades of research grounded in the theory of planned behavior (Ajzen, 1991) and growing recognition of sophisticated nudging approaches (Thaler & Sunstein, 2008), our analysis demonstrates that 71% of interventions still rely primarily on communication campaigns. This overreliance on information provision persists even as our findings confirm a substantial knowledge-behavior gap, with multiple studies reporting increased awareness without corresponding behavioral changes. The predominance of traditional approaches suggests that intervention designers either lack awareness of behavioral science advances or face systematic constraints that prevent implementation of more sophisticated strategies.

The effectiveness analysis provides sobering evidence about intervention sustainability. Only 12.5% of studies achieved high effectiveness >50% improvement), while a concerning 19% showed mixed or declining results over time. This pattern aligns with Nisa et al. (2019) observation about questionable long-term sustainability, but our analysis extends this concern by identifying specific mechanisms of failure. Xu et al. (2023) documented a particularly troubling phenomenon: motivational crowding-out, where economic incentives increased recycling behavior while simultaneously reducing intrinsic environmental motivation by 36%. This finding challenges the widespread assumption that any increase in recycling behavior represents progress, suggesting instead that intervention design must consider both immediate behavioral outcomes and long-term motivational sustainability.

Furthermore, the synthesized literature reveals a spectrum of academic perspectives on recycling behavior that sometimes conflict. Engineering and operations studies treat recycling as primarily a function of physical convenience and system optimization; behavioral economics emphasizes rational financial levers, though our synthesis reveals this to be polarizing—some studies document dramatic gains through economic incentives while others warn of motivational crowding-out; and environmental psychology focuses on social norms, community identity and cultural traditions. Together, these perspectives demonstrate that the optimal intervention is highly paradigm-dependent, underscoring the need to move beyond siloed disciplinary approaches toward transdisciplinary intervention design.

6.1 Multi-level barrier interactions and system complexity

Our barrier analysis reveals that implementation challenges operate through complex multi-level interactions rather than isolated constraints. While previous reviews have identified barriers to individual recycling behaviors (Knickmeyer, 2020; Zhang, 2023), our focus on intervention effectiveness exposes a different pattern: meso and micro barriers (131 of 150 mentions) dominate over macro-level constraints. This distribution suggests that even well-designed policies fail primarily through implementation breakdowns and individual participation constraints rather than systemic policy gaps.

The prevalence of practical implementation issues ($n = 28$) as the most frequently reported barrier category highlights a critical oversight in intervention design. These failures, ranging from missed collections (Cotterill et al., 2009) to staff rudeness (Dutta-Powell et al., 2023), represent preventable breakdowns that undermine otherwise sound interventions. This finding resonates with implementation science literature suggesting that intervention failure often results from inadequate attention to delivery mechanisms rather than flawed underlying theory (Damschroder et al., 2009).

Infrastructure and facility barriers ($n = 24$) emerged as particularly influential, with six of eight high-effectiveness studies incorporating infrastructure modifications as core components. This pattern suggests

a hierarchy of intervention prerequisities: without addressing basic structural barriers, behavioral interventions may be fundamentally limited in their potential impact. The technology adoption resistance documented by Löytty et al. (2023), including 'big brother aversion' toward IoT composting systems, further illustrates how infrastructure barriers extend beyond physical constraints to encompass technological acceptance challenges that intervention designers must anticipate.

6.2 Cultural context and the limits of universal solutions

The geographic concentration of studies (78% from Asia and Europe) raises fundamental questions about the generalizability or recycling intervention knowledge. Recycling practices vary dramatically across contexts: from Sweden's systematic cardboard removal to informal waste collector communities in developing nations. These variations reflect not merely different levels of infrastructure development but fundamentally different conceptualizations of waste, value and responsibility.

The three types of fatalism identified by Entwistle (2023); John and Stoker (2019) provide a useful framework for understanding differential intervention responses. Passive fatalists who 'let it be' may explain the persistent 20–30% of populations that remain unresponsive to any intervention type. Protective fatalists seeking shelter from perceived threats might respond better to interventions that emphasize risk mitigation rather than environmental benefits. Pathological fatalists who deliberately oppose system requirements present particular challenges, potentially explaining some instances of intervention backfire effects. This typology suggests that universal intervention approaches are inherently limited and that segmented strategies targeting different motivational profiles may be necessary.

6.3 The knowledge-practice translation challenge

A striking finding emerges from comparing intervention types with effectiveness outcomes: sophisticated behavioral interventions based on robust theoretical foundations (social norms 3%; performance-based messaging 6%) remain dramatically underutilized despite evidence of effectiveness when properly implemented. This knowledge-practice translation gap parallels challenges identified in healthcare implementation, where evidence-based practices often take 17 years to reach routine clinical practice (Balas & Boren, 2000).

Several factors may explain this translation failure. First, the complexity of behavioral interventions may exceed the implementation capacity of a typical waste management organization. Second, the immediate visibility of infrastructure changes may make them more appealing to policymakers than subtle behavioral nudges with delayed effects. Third, the predominance of engineering and policy perspectives in waste management may create blind spots regarding behavioral science insights. This suggests a need for transdisciplinary collaboration in intervention design, bringing together waste management practitioners, behavioral scientists and implementation researchers.

Our analysis reveals a fundamental limitation in how recycling intervention success is measured. The predominance of self-reported measures (54% of studies) and focus on quantitative metrics like tonnage or participation rates may obscure important qualitative dimensions of success. For instance, interventions that successfully build community engagement or environmental consciousness might be deemed failures if they don't immediately increase recycling rates, yet these outcomes might be essential precursors to long-term behavioral change.

6.4 Implications for policy and practice

The evidence synthesized here suggests several critical implications for recycling intervention design. First, single-component interventions appear insufficient for achieving substantial behavioral change; the most successful interventions combined multiple elements addressing different barrier levels simultaneously. Second, infrastructure modifications emerge as potentially necessary (though not sufficient) conditions for intervention success, suggesting that behavioral approaches cannot compensate for fundamental system inadequacies. Third, the temporal dynamics of intervention effectiveness, with many showing declining

effects over time, indicate that sustained behavioral change requires ongoing intervention adaptation rather than one-time implementation.

Practically, this means policymakers must shift from linear, information-first strategies to a comprehensive, systems-thinking approach. Our synthesis implies that before launching any behavioral or communication campaign, municipalities must first conduct rigorous 'barrier audits' to ensure meso-level infrastructure (e.g. bin proximity, reliable collection schedules) is fully optimized. If micro-level convenience and space constraints remain unaddressed, our results indicate that even the most sophisticated nudges or economic incentives will fall flat, or worse, lead to motivational crowding-out. Consequently, we recommend that intervention funding be systematically reallocated away from standalone awareness campaigns. Instead, resources should be directed toward co-designed, community-specific pilot programs that couple tangible infrastructural upgrades with targeted, performance-based messaging and social norm interventions, thereby addressing the physical and psychological dimensions of recycling simultaneously.

Translating these findings into practice requires concrete action at each level of the delivery chain. *Municipalities* should prioritize infrastructure audits before launching communication campaigns—ensuring adequate bin provision, accessible collection points and reliable collection schedules—since our results indicate that behavioral interventions applied on top of unresolved meso-level barriers are unlikely to achieve lasting change. *Waste service providers* should invest in staff training and service quality monitoring, given that staff behavior and missed collections emerged as among the most frequently cited barriers to intervention effectiveness. *Designers of household recycling programs* should adopt a staged approach: first resolving practical and infrastructural barriers, then deploying evidence-based behavioral tools such as performance-based feedback and social norm messaging rather than generic awareness campaigns. Finally, all programs should build in sustainability mechanisms from the outset—For instance, planning an explicit transition from extrinsic economic incentives to intrinsic motivation-building strategies—to guard against the motivational crowding-out documented in this review.

The underrepresentation of studies from the Global South, where waste management challenges are most acute and growing rapidly (United Nations Environment Programme & International Solid Waste Association, 2024), represents a critical knowledge gap. With global waste projected to increase from 2.1 to 3.8 billion tons by 2050, primarily in developing regions, the field urgently needs research examining intervention effectiveness in low-resource, informal waste management contexts.

6.5 Limitations

This review faces several limitations that qualify our conclusions. First, the predominance of studies from developed economies limits generalizability to contexts where waste management challenges are most pressing. Second, our focus on household recycling excludes potentially relevant insights from commercial and industrial waste management interventions. Third, following standard PRISMA scoping procedures, our search strategy was restricted to titles, abstracts and keywords to ensure thematic relevance. Consequently, studies that may have encountered recycling barriers peripherally but did not highlight them as primary findings in their core metadata were excluded from this synthesis. Finally, the rapid evolution of waste management technologies and policies means that some included studies may not reflect current best practices. A further limitation is that this review's conclusions about 'intervention failure' are partly inferred from primary studies originally designed to measure behavioral outcomes rather than intervention architecture; the strength of these conclusions therefore depends on the quality and completeness of implementation reporting in those underlying studies.

6.6 Future research directions

This review identifies several critical, highly specific areas requiring future investigation to bridge the knowledge-practice gap. First, given the alarming rate of declining intervention effects and instances of motivational crowding-out identified in our synthesis, future research must prioritize longitudinal experimental designs. Specifically, researchers should test dynamic, adaptive interventions that systematically shift from extrinsic economic rewards to intrinsic motivational and social norm builders over time to

sustain behavioral change. Second, to address the severe geographic imbalance, studies in the Global South must move beyond basic observational research. Future work should apply the macro-meso-micro barrier framework to evaluate how interventions designed for formal waste systems can be adapted for informal waste management economies, specifically addressing the unique infrastructural constraints and cultural norms of these regions. Third, while this review focused on recycling, future empirical studies should directly test if and how recycling interventions cause positive or negative spillover effects onto upstream waste prevention behaviors (e.g. reducing household consumption), thus aligning with the broader waste hierarchy priorities. Finally, addressing the predominant meso-level practical implementation failures ($n = 28$) identified in this review requires a shift toward transdisciplinary action research. Scholars must partner directly with municipal waste authorities and engage citizens as active stakeholders to co-develop and test localized implementation toolkits and decision-support systems. This participatory approach will ensure that theoretically sound behavioral nudges are actually feasible within the operational constraints of real-world waste management. Furthermore, as the reporting of intervention metrics and outcome measurements becomes more standardized across the field, future research should aim to validate the findings of this narrative synthesis by conducting a formal quantitative meta-analysis.

7 Conclusions

This systematic review reveals a critical disconnect between recycling intervention design and effectiveness, with 71% of interventions relying on communication campaigns despite persistent evidence of the knowledge-behavior gap. Only 12.5% of interventions achieved high effectiveness (>50% improvement), while 19% showed declining effects over time, highlighting fundamental sustainability challenges in current approaches. The predominance of meso and micro-level barriers (87% of 150 identified constraints) demonstrates that intervention failures stem primarily from implementation breakdowns and household participation constraints rather than policy gaps, with practical implementation issues ($n = 28$) and infrastructure inadequacies ($n = 24$) representing the most frequent obstacles. Notably, six of eight high-effectiveness interventions incorporated infrastructure modifications as core components, suggesting that addressing structural barriers may be a necessary condition for behavioral change. The persistent underutilization of evidence-based behavioral interventions (social norms, performance-based messaging) despite their theoretical support indicates a substantial knowledge-practice translation gap in the field.

These findings suggest that effective recycling interventions require multi-level approaches that simultaneously address infrastructure, implementation quality and behavioral dimensions, while the geographic concentration of research in developed economies raises critical questions about the applicability of current knowledge to developing regions where waste management challenges are most acute and rapidly growing

Author contributions

CRediT: **Sami Heikkinen:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Validation, Writing – original draft, Writing – review & editing; **Kaisu Isomäki:** Conceptualization, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing; **Taina Salakka-Kontunen:** Formal analysis, Validation, Writing – original draft, Writing – review & editing.

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Registration and protocol

This systematic review was not registered, and a protocol was not prepared in advance of the review.

Declaration of generative AI

During the preparation of this work, the author(s) used Grammarly and Gemini to polish the language used in the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Ajzen, I. (1991, Dec). The theory of planned behavior. *Organ Behav Hum Decis Process*, *50*(2), 179–211. https://doi.org/10.1016/0749-5978(91)90020-T

Ajzen, I. (2014). The theory of planned behavior, *Handbook of Theories of Social Psychology: Volume 1* (pp. 438–59). 1 Oliver's Yard, 55 City Road, London EC1Y 1SP United Kingdom: SAGE Publications Ltd. https://doi.org/10.4135/9781446249215.n22

Atthirawong, W. (2016). Factors Affecting Household Participation In Solid Waste Management Segregation And Recycling In Bangkok, Thailand. In: ECMS 2016 Proceedings edited by Thorsten Claus, Frank Herrmann, Michael Manitz, Oliver Rose. ECMS. p. 198–203 https://doi.org/10.7148/2016-0198

Awino, F. B., & Apitz, S. E. (2024, Jan). Solid waste management in the context of the waste hierarchy and circular economy frameworks: An international critical review. *Integr Environ Assess Manag*, *20*(1), 9–35. https://doi.org/10.1002/ieam.4774

Balas, E. A., & Boren, S. A. (2000, Aug). Managing clinical knowledge for health care improvement. *Yearb Med Inform*, *09*(01), 65–70. https://doi.org/10.1055/s-0038-1637943

Bernstad, A., La Cour Jansen, J., & Aspegren, A. (2013). Door-stepping as a strategy for improved food waste recycling behaviour-evaluation of a full-scale experiment. *Resources, Conservation and Recycling*, *73*, 94–103. https://doi.org/10.1016/j.resconrec.2012.12.012

Bian, S., Xue, Y., Li, C., Wu, W., Ma, Z., Okoro, J. O., & Harder, M. K. (2022, 01). Depositing waste in "certain place" at "certain time": Does this policy improve recycling performance?. *Resour Conserv Recycl*, *176*, 105935. https://doi.org/10.1016/j.resconrec.2021.105935

Bortoleto, A. P., & Hanaki, K. (2007, 06). Report: Citizen participation as a part of integrated solid waste management: Porto alegre case. *Waste Manag Res*, *25*(3), 276–82. https://doi.org/10.1177/0734242X07079204

Bowman, N., Goodwin, J., Jones, P., & Weaver, N. (1998, 12). Sustaining recycling: Identification and application of limiting factors in kerbside recycling areas. *Int J Sustainable Dev World Ecol*, *5*(4), 263–76. https://doi.org/10.1080/13504509809469991

Bramer, W. M., Rethlefsen, M. L., Kleijnen, J., & Franco, O. H. (2017, Dec 6). Optimal database combinations for literature searches in systematic reviews: A prospective exploratory study. *Syst Rev*, *6*(1), 245. https://doi.org/10.1186/s13643-017-0644-y

Bryce, W. J., Day, R., & Olney, T. J. (1997, 06). Commitment approach to motivating community recycling: New Zealand curbside trial. *J Consum Aff*, *31*(1), 27–52. https://doi.org/10.1111/j.1745-6606.1997.tb00825.x

Caldera, S., Jayasinghe, R., Desha, C., Dawes, L., & Ferguson, S. (2022). Evaluating barriers, enablers and opportunities for closing the loop through "Waste Upcycling": A systematic literature review. *Journal of Sustainable Development of Energy, Water and Environment Systems*, *10*(1), 1–20. https://doi.org/10.13044/j.sdewes.d8.0367] – [

Chase, N. L., Dominick, G. M., Trepal, A., Bailey, L. S., & Friedman, D. B. (2009). This is public health: Recycling counts!" description of a pilot health communications campaign. *International Journal of Environmental Research and Public Health*, *6*, 2980–91. https://doi.org/10.3390/ijerph6122980

Chen, F., Chen, W., Hou, J., & Li, W. (2022, 02). Research on the variations in individual waste separation behavior due to different information strategies–Mediating effects of psychological distance. *J Environ Manage*, *304*, 114320. https://doi.org/10.1016/j.jenvman.2021.114320

Cole, C., Quddus, M., Wheatley, A., Osmani, M., & Kay, K. (2014, 02). The impact of local authorities' interventions on household waste collection: A case study approach using time series modelling. *Waste Manag*, *34*(2), 266–72. https://doi.org/10.1016/j.wasman.2013.10.018

Cotterill, S., John, P., Liu, H., & Nomura, H. (2009, 11). Mobilizing citizen effort to enhance environmental outcomes: A randomized controlled trial of a door-to-door recycling campaign. *J Environ Manage*, *91*(2), 403–10. https://doi.org/10.1016/j.jenvman.2009.09.010

Cox, J., Giorgi, S., Sharp, V., Strange, K., Wilson, D. C., & Blakey, N. (2010, Mar). Household waste prevention--a review of evidence. *Waste Manag Res*, *28*(3), 193–219. https://doi.org/10.1177/0734242X10361506

Czajkowski, M., Zagórska, K., & Hanley, N. (2019, 11). Social norm nudging and preferences for household recycling. *Res Energy Econ*, *58*, 101110. https://doi.org/10.1016/j.reseneeco.2019.07.004

Damschroder, L. J., Aron, D. C., Keith, R. E., Kirsh, S. R., Alexander, J. A., & Lowery, J. C. (2009, Aug 7). Fostering implementation of health services research findings into practice: A consolidated framework for advancing implementation science. *Implement Sci*, *4*(1), 50. https://doi.org/10.1186/1748-5908-4-50

De Fano, D., Schena, R., & Russo, A. (2022, 07). Empowering plastic recycling: Empirical investigation on the influence of social media on consumer behavior. *Resour Conserv Recycl*, *182*, 106269. https://doi.org/10.1016/j.resconrec.2022.106269

De Gisi, S., Casella, P., Sabia, G., Farina, R., Landolfo, P., Notarnicola, M., & De Feo, G. (2017, 10). Assessing the public perception of islanders regarding the implementation of new technologies to optimize the municipal solid waste management system: A Mediterranean case study. *J Clean Prod*, *164*, 1586–601. https://doi.org/10.1016/j.jclepro.2017.07.090

Del Cimmuto, A., Mannocci, A., Ribatti, D., Boccia, A., & La Torre, G. (2014, 06). Impact on knowledge and behaviour of the general population of two different methods of solid waste management: An explorative cross-sectional study. *Waste Manag Res*, *32*(6), 556–61. https://doi.org/10.1177/0734242X14536461

Dhokhikah, Y., Trihadiningrum, Y., & Sunaryo, S. (2015, 09). Community participation in household solid waste reduction in surabaya, Indonesia. *Resour Conserv Recycl*, *102*, 153–62. https://doi.org/10.1016/j.resconrec.2015.06.013

Díaz, M. J., Martínez, E., Piñeiro, C., Palavecinos, M., Benayas, J., & Toribio, M. A. (2012, 07). Involvement of citizens in hazardous waste management and use of recycling centres in the city of Madrid (Spain). *Waste Manag Res*, *30*(7), 689–99. https://doi.org/10.1177/0734242X11432189

Dutta-Powell, R., Court, C., & Clark, A. (2023, 12). Interventions for effective waste management and segregation: Evidence from a randomised controlled trial in the changzamtog district of Thimphu, Bhutan. *Cleaner Waste Systems*, *6*, 100121. https://doi.org/10.1016/j.clwas.2023.100121

Entwistle, T. (2023). Why nudge sometimes fails: Fatalism and the problem of behaviour change, *Beyond Nudge*, 88–107. Policy Press. https://doi.org/10.1332/policypress/9781447369141.003.0005

Environment [Internet]. (n.d.). [cited 2025 Oct 8]. Waste framework directive. Available from: https://environment.ec.europa.eu/topics/waste-and-recycling/waste-framework-directive_en

Flygansvær, B., Samuelsen, A. G., & Støyle, R. V. (2021, Oct 8). The power of nudging: How adaptations in reverse logistics systems can improve end-consumer recycling behavior. *Int J Phys Distrib Logist Manag*, *51*(9), 958–77. https://doi.org/10.1108/IJPDLM-12-2020-0389

Geislar, S. (2017, 10). The new norms of food waste at the curb: Evidence-based policy tools to address benefits and barriers. *Waste Manag*, *68*, 571–80. https://doi.org/10.1016/j.wasman.2017.07.010

GOV.UK [Internet]. (2011). [cited 2025 Oct 8]. Securing the future: delivering UK sustainable development strategy. Available from: https://www.gov.uk/government/publications/securing-the-future-delivering-uk-sustainable-development-strategy

Heidbreder, L. M., Bablok, I., Drews, S., & Menzel, C. (2019, Jun). Tackling the plastic problem: A review on perceptions, behaviors, and interventions. *Sci Total Environ*, *668*, 1077–93. https://doi.org/10.1016/j.scitotenv.2019.02.437

Hernández, O., Rawlins, B., & Schwartz, R. (1999, Oct 1). Voluntary recycling in Quito: Factors associated with participation in a pilot programme. *Environ Urban*, *11*(2), 145–59. https://doi.org/10.1630/095624799101284940

Higgins, JPT, Thomas, J, Chandler, J, Cumpston, M, Li, T, Page, MJ, & Welch, V. (2019). Cochrane handbook for systematic reviews of interventions.

Hotta, Y., & Aoki-Suzuki, C. (2014, 09). Waste reduction and recycling initiatives in Japanese cities: Lessons from yokohama and kamakura. *Waste Manag Res*, *32*(9), 857–66. https://doi.org/10.1177/0734242X14539721

Huang, Y. Y., Tamas, P. A., & Harder, M. K. (2018, 03). Information with a smile – does it increase recycling? *J Clean Prod*, *178*, 947–53. https://doi.org/10.1016/j.jclepro.2018.01.006

John, P., & Stoker, G. (2019, Apr). Rethinking the role of experts and expertise in behavioural public policy. *Policy Polit*, *47*(2), 209–25. https://doi.org/10.1332/030557319X15526371698257

Karout, N., & Altuwaijri, S. (2012, Jul 1). Impact of health education on community knowledge, attitudes and behaviour towards solid waste management in al ghobeiry, Beirut. *East Mediterr Health J*, *18*(7), 777–85. https://doi.org/10.26719/2012.18.7.777

Kattoua, M. G., Al-Khatib, I. A., & Kontogianni, S. (2019, 7). Barriers on the propagation of household solid waste recycling practices in developing countries: State of palestine example. *J Mater Cycles Waste Manag*, *21*(4), 774–85. https://doi.org/10.1007/s10163-019-00833-5

Kaza, S., Shrikanth, S., & Chaudhary, S. (2021, Jul). *More Growth, Less Garbage [Internet]*. World Bank. [cited 2025 Oct 8]. Available from: https://openknowledge.worldbank.org/handle/10986/35998

Kazerooni Sadi, M. A., Abdullah, A., Navazandeh Sajoudi, M., Kamal, M. F. M., Torshizi, F., & Taherkhani, R. (2012, Jan). Reduce, reuse, recycle and recovery in sustainable construction waste management. *Adv Mat Res*, *446-449*, 937–44. https://doi.org/10.4028/www.scientific.net/AMR.446-449.937

Knickmeyer, D. (2020, Feb). Social factors influencing household waste separation: A literature review on good practices to improve the recycling performance of urban areas. *J Clean Prod*, *245*, 118605118605. https://doi.org/10.1016/j.jclepro.2019.118605

Komatsu, H., Kubota, H., Tanaka, N., Griffin, M., Link, J., Geher, G., Fisher, M. L., & Xin, B. (2022, Nov 3). Cross-cultural comparison of nudging effects for environmental protection: A case-study of risk-averse attitudes toward disposable plastics. *PLoS One*, *17*(11), e0277183. https://doi.org/10.1371/journal.pone.0277183

Kutralam-Muniasamy, G., Shruti, V. C., & Pérez-Guevara, F. (2023, 10). Citizen involvement in reducing end-of-life product waste in Mexico City. *Sustain Prod Consum*, *41*, 167–78. https://doi.org/10.1016/j.spc.2023.08.010

Lakhan, C. (2016, 03). Out of sight, out of mind: Issues and obstacles to recycling in ontario's multi residential buildings. *Resour Conserv Recycl*, *108*, 1–9. https://doi.org/10.1016/j.resconrec.2016.01.005

Li, S. (2003). Recycling behavior under China's social and economic transition: The case of metropolitan wuhan. *Environment and Behavior*, *35*, 784–801. https://doi.org/10.1177/0013916503254819

Li, C., Wang, Y., Li, Y., Huang, Y., & Harder, M. K. (2021). The incentives May not be the incentive: A field experiment in recycling of residential food waste. *Resources, Conservation and Recycling*, *168*, 105316. https://doi.org/10.1016/j.resconrec.2020.105316

Liang, Y., Li, Z., Feng, S., & Zhang, Y. (2024, 05). Can we "Nudge" people to better waste separation behaviours? Policy interventions mediated by habit, sense of separation efficiency and external environmental perceptions. *Waste Manag Res*, *42*(5), 372–83. https://doi.org/10.1177/0734242X231187579

Linder, N., Lindahl, T., & Borgström, S. (2018, Mar 22). Using behavioural insights to promote food waste recycling in urban Households—Evidence from a longitudinal field experiment. *Front Psychol*, *9*, 352. https://doi.org/10.3389/fpsyg.2018.00352

Ling, M., Xu, L., & Chu, X. (2023, 03). Heterogeneous effects of other-regarding interventions on household recycling: A field experimental study. *J Environ Manage*, *329*, 117102. https://doi.org/10.1016/j.jenvman.2022.117102

Liu, Y., Zou, C., Tai, J., & Che, Y. (2023, 08). Evaluation of the household solid waste mandatory sorting policy in shanghai: A 1-year public opinion survey. *Waste Manag Res*, *41*(8), 1305–17. https://doi.org/10.1177/0734242X231155097

Löytty, T., Rantamäki, S., Fontell, H., & Karlsson, K. (2023, Sep 30). Iot-sensor-equipped food waste bio-composter to households and to advance egovernment in municipality authorities' waste management practices. *Agris On-line Pap Econ Inform*, *15*(3), 105–18. https://doi.org/10.7160/aol.2023.150309

Macklin, J., Curtis, J., & Smith, L. (2023, Oct). Interdisciplinary, systematic review found influences on household recycling behaviour are many and multifaceted, requiring a multi-level approach. *Resour Conserv Recycl Adv*, *18*, 200152200152. https://doi.org/10.1016/j.rcradv.2023.200152

Manomaivibool, P., Srivichai, M., Unroj, P., & Dokmaingam, P. (2018, 09). Chiang rai zero waste: Participatory action research to promote source separation in rural areas. *Resour Conserv Recycl*, *136*, 142–52. https://doi.org/10.1016/j.resconrec.2018.04.002

Marquez, C. T., & Gomez, B. (2022). Reduce, reuse, recycle. In WL Filho Azul, AM Brandli, & G Özuar (Eds.), 626–34). Consumption R, Production.

McHugh, M. L. (2012). Interrater reliability: The kappa statistic. *Biochem Med (Zagreb)*, *22*(3), 276–82. https://doi.org/10.11613/BM.2012.031

Meen-Chee, H., & Narayanan, S. (2006). Restoring the shine to a pearl: Recycling behaviour in penang, Malaysia. *Dev Change*, *37*(5), 1117–36. https://doi.org/10.1111/j.1467-7660.2006.00515.x

Mehner, E., Naidoo, A., Hellwig, C., Bolton, K., & Rousta, K. (2020). The influence of user-adapted, instructive information on participation in a recycling scheme: A case study in a medium-sized Swedish city. *Recycling*, *5*(2), 7. https://doi.org/10.3390/recycling5020007

Methley, A. M., Campbell, S., Chew-Graham, C., McNally, R., & Cheraghi-Sohi, S. (2014, Nov 21). PICO, PICOS and SPIDER: A comparison study of specificity and sensitivity in three search tools for qualitative systematic reviews. *BMC Health Serv Res*, *14*(1), 579. https://doi.org/10.1186/s12913-014-0579-0

Moeini, B., Ayubi, E., Barati, M., Bashirian, S., Tapak, L., Ezzati-Rastgar, K., & Hashemian, M. (2023). Effect of household interventions on promoting waste segregation behavior at source: A systematic review. *Sustainability*, *15*(24), 16546. https://doi.org/10.3390/su152416546

Munn, Z., Peters, M. D. J., Stern, C., Tufanaru, C., McArthur, A., & Aromataris, E. (2018). Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Med Res Methodol [Internet]*, *18*(1), 143. Available from: https://doi.org/10.1186/s12874-018-0611-x

Nathanson, A. J. (2025). solid-waste management. In: Encyclopedia Britannica [Internet]. Available from: https://www.britannica.com/technology/solid-waste-management

Nepal, M., Karki Nepal, A., Khadayat, M. S., Rai, R. K., Shyamsundar, P., & Somanathan, E. (2023). Low-cost strategies to improve municipal solid waste management in developing countries: Experimental evidence from Nepal. *Environ Resour Econ*, 84(3), 729–752. https://doi.org/10.1007/s10640-021-00640-3

Nguyen, T. K. N. (2022). Individuals' waste separation practice in a relationship with social bonds: A case study of Hanoi, Vietnam. *Int J Sociol Soc Policy*, *42*(9/10), 816–30. https://doi.org/10.1108/IJSSP-07-2021-0184

Nisa, C. F., Bélanger, J. J., Schumpe, B. M., & Faller, D. G. (2019). Meta-analysis of randomised controlled trials testing behavioural interventions to promote household action on climate change. *Nat Commun*, *10*(1), 4545. https://doi.org/10.1038/s41467-019-12457-2

Nowakowski, P., Kuśnierz, S., Płoszaj, J., & Sosna, P. (2021). Collecting small-waste electrical and electronic equipment in Poland—How can containers help in disposal of E-Waste by individuals?. *Sustainability*, *13*(22), 12422. https://doi.org/10.3390/su132212422

Okonta, F. N., & Mohlalifi, M. (2020, 03). Assessment of factors affecting source recycling among metropolitan johannesburg residents. *Waste Manag*, *105*, 445–9. https://doi.org/10.1016/j.wasman.2020.02.006

Omran, A., Mahmood, A., Aziz, H., & Robinson, G. M. (2009). Investigating households attitude toward recycling of solid waste in Malaysia: A case study. *International Journal of Environmental Research*, *3*, 275–88.

Otto, S., Kibbe, A., Henn, L., Hentschke, L., & Kaiser, F. G. (2018). The economy of E-waste collection at the individual level: A practice oriented approach of categorizing determinants of E-waste collection into behavioral costs and motivation. *J Clean Prod*, *204*, 33–40. https://doi.org/10.1016/j.jclepro.2018.08.293

Page, M, McKenzie, J, Bossuyt, P, Boutron, I, Hoffmann, T, Mulrow, C, Shamseer, L, Tetzlaff, J, Akl, E, Brennan, S, Chou, R, Glanville, J, Grimshaw, J, Hróbjartsson, A, Lalu, M, Li, T, Loder, E, Mayo-Wilson, E, McDonald, S, McGuinness, L, Stewart, L, Thomas, J, Tricco, A, Welch, V, Whiting, P, & Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ [Internet]*, 372, 372. [cited 2022 May 26]. Available from: https://doi.org/10.1136/BMJ.N71

Parajuly, K., Fitzpatrick, C., Muldoon, O., & Kuehr, R. (2020). Behavioral change for the circular economy: A review with focus on electronic waste management in the EU. *Resources, Conservation & Recycling: X*, *6*(100035), 100035.

Park, S., & Lah, T. J. (2015). Analyzing the success of the volume-based waste fee system in South Korea. *Waste Manag*, *43*, 533–8. https://doi.org/10.1016/j.wasman.2015.06.011

Passafaro, P., & Livi, S. (2017). Comparing determinants of perceived and actual recycling skills: The role of motivational, behavioral and dispositional factors. *J Environ Educ*, *48*(5), 347–56. https://doi.org/10.1080/00958964.2017.1320961

Pegels, A., Castañeda, J. L., Humphreys, C., Kötter, C., Negre, M., Weidner, C., & Kutzner, F. (2022, 02). Aligning recycling behaviors and the recycling system – towards a full cycle of materials and behavioral methods. *Waste Manag*, *138*, 1–7. https://doi.org/10.1016/j.wasman.2021.11.021

Pongpunpurt, P., Muensitthiroj, P., Pinitjitsamut, P., Chuenchum, P., Painmanakul, P., Chawaloesphonsiya, N., & Poyai, T. (2022). Studying waste separation behaviors and environmental impacts toward sustainable solid waste management: A case study of bang chalong housing, samut prakan, Thailand. *Sustainability*, *14*(9), 5040. https://doi.org/10.3390/su14095040

Pongpunpurt, P., Navamajiti, N., Sriroongvikrai, K., Onnom, M., Pinitjitsamut, P., Painmanakul, P., Chawaloesphonsiya, N., & Poyai, T. (2021). Analyzing productivity and behavior of plastic drop-off points: A case study of send plastic home project in plastic waste recycling during COVID-19 outbreak. *Warasan Witsawakamsat, Chulalongkorn Univ*, *25*(10), 1–11. https://doi.org/10.4186/ej.2021.25.10.1

Popay, J., Roberts, H., Sowden, A., Petticrew, M., Arai, L., & Rodgers, M., et al. (2006). *Guidance on the conduct of narrative synthesis in systematic reviews: A product from the ESRC Methods Programme*. Lancaster University. Lancaster, England.

Pullin, A. S., Frampton, G. K., Livoreil, B., & Petrokofsky, G. (2022). Guidelines and Standards for Evidence Synthesis in Environmental Management.

Read, A. D. (1999). A weekly doorstep recycling collection, I had no idea we could!': Overcoming the local barriers to participation. *Resources, Conservation and Recycling*, *26*, 217–49. https://doi.org/10.1016/S0921-3449(99)00008-7

Read, M., Gregory, M. K., & Phillips, P. S. (2009). An evaluation of four key methods for monitoring household waste prevention campaigns in the UK. *Resour Conserv Recycl*, *54*(1), 9–20. https://doi.org/10.1016/j.resconrec.2009.05.004

Remr, J. (2018). Crowding-out effect of financial incentives for households to recycle waste. In Seville, Spain; 2018. p. 269–79.

Roe, B. E., Qi, D., Beyl, R. A., Neubig, K. E., Apolzan, J. W., & Martin, C. K. (2022). A randomized controlled trial to address consumer food waste with a technology-aided tailored sustainability intervention. *Resour Conserv Recycl*, *179*, 106121. https://doi.org/10.1016/j.resconrec.2021.106121

Rousta, K., Bolton, K., Lundin, M., & Dahlén, L. (2015). Quantitative assessment of distance to collection point and improved sorting information on source separation of household waste. *Waste Manag*, *40*, 22–30. https://doi.org/10.1016/j.wasman.2015.03.005

Roy, D., Berry, E., & Dempster, M. (2022). If it is not made easy for me, I will just not bother". A qualitative exploration of the barriers and facilitators to recycling plastics. *PLoS One*, *17*(5), e0267284. https://doi.org/10.1371/journal.pone.0267284

Sabbir, M. M., Taufique, K. M. R., & Nomi, M. (2023). Consumers' reverse exchange behavior and e-waste recycling to promote sustainable post-consumption behavior. *Asia Pac J Mark Logist*, *35*(10), 2484–500. https://doi.org/10.1108/APJML-07-2022-0647

Schmidt, W., Haider, I., Hussain, M., Safdar, M., Mustafa, F., Massey, T., Angelo, G., Williams, M., Gower, R., Hasan, Z., Sharma Waddington, H., Anjum, N., & Biran, A. (2022). The effect of improving solid waste collection on waste disposal behaviour and exposure to environmental risk factors in urban low-income communities in Pakistan. *Trop Med Int Health*, *27*(7), 606–18. https://doi.org/10.1111/tmi.13787

Sewak, A., Kim, J., Rundle-Thiele, S., & Deshpande, S. (2021). Influencing household-level waste-sorting and composting behaviour: What works? A systematic review (1995–2020) of waste management interventions. *Waste Manag Res*, *39*(7), 892–909. https://doi.org/10.1177/0734242X20985608

Shan, X., Ang, W. L., & Yang, E. H. (2020). Mobile app-aided risks, attitudes, norms, abilities and self-regulation (RANAS) approach for recycling behavioral change in Singapore. *Resour Conserv Recycl*, *162*, 105049. https://doi.org/10.1016/j.resconrec.2020.105049

Siepelmeyer, H., & Otterbring, T. (2022). Citizen coherence and cultivated cleanliness: Using technology-induced social norms to strengthen sustainable household bonds. *Front Sustain*, *3*. 899938. https://doi.org/10.3389/frsus.2022.899938

Sintov, N., Geislar, S., & White, L. V. (2019). Cognitive accessibility as a new factor in proenvironmental spillover: Results from a field study of household food waste management. *Environ Behav*, *51*(1), 50–80. https://doi.org/10.1177/0013916517735638

Stoknes, P. E. (2015). *What we think about when we try not to think about global warming: toward a new psychology of climate action*. Chelsea Green Publishing. White River Junction, Vermont.

Takahashi, M., & Kang, N. (2020). Proposal and kansei evaluation of a garbage can to change human behaviour based on fun theory, Advances in Industrial Design (pp. 717–24). Cham: Springer International Publishing. (Advances in intelligent systems and computing). https://doi.org/10.1007/978-3-030-51194-4_93

Thaler, R. H., & Sunstein, C. R. (2008). *Nudge: Improving decisions about health, wealth, and happiness*. 293. Yale University Press.

The Editors of Encyclopedia Britannica. (2025). recycling. In: Encyclopedia Britannica [Internet]. Available from: https://www.britannica.com/science/recycling

Timlett, R. E., & Williams, I. D. (2008, 2). Public participation and recycling performance in england: A comparison of tools for behaviour change. *Resour Conserv Recycl*, *52*(4), 622–34. https://doi.org/10.1016/j.resconrec.2007.08.003

Tong, X., Nikolic, I., Dijkhuizen, B., Van Den Hoven, M., Minderhoud, M., Wäckerlin, N., Wang, T., & Tao, D. (2018, 06). Behaviour change in post-consumer recycling: Applying agent-based modelling in social experiment. *J Clean Prod*, *187*, 1006–13. https://doi.org/10.1016/j.jclepro.2018.03.261

United Nations Environment Programme, International Solid Waste Association (2024). Global Waste Management Outlook 2024 - Beyond an age of waste: Turning rubbish into a resource. United Nations Environment Programme.

Varotto, A., & Spagnolli, A. (2017, Aug). Psychological strategies to promote household recycling. A systematic review with meta-analysis of validated field interventions. *J Environ Psychol*, *51*, 168–88. https://doi.org/10.1016/j.jenvp.2017.03.011

Vining, J., & Ebreo, A. (1989, 01). An evaluation of the public response to a community recycling education program. *Soc Nat Resour*, *2*(1), 23–36. https://doi.org/10.1080/08941928909380673

Vining, J., & Ebreo, A. (1990). What makes a recycler? A comparison of recyclers and non-recyclers. *Environmental Behaviour*, *22*, 55–73. https://doi.org/10.1177/0013916590221003

Wakefield, F. (2022). Wakefield F. World Economic Forum. [cited 2025 Oct 8]. Top 25 recycling facts and statistics for 2022. Available from: https://www.weforum.org/stories/2022/06/recycling-global-statistics-facts-plastic-paper/

Waste hierarchy [Internet]. (n.d.). [cited 2025 Oct 8]. Available from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=LEGISSUM:waste_hierarchy

Welfens, M. J., Nordmann, J., & Seibt, A. (2016, Sep). Drivers and barriers to return and recycling of mobile phones. Case studies of communication and collection campaigns. *J Clean Prod*, *132*, 108–21. https://doi.org/10.1016/j.jclepro.2015.11.082

Willman, K. W. (2015, 11). Information sharing and curbside recycling: A pilot study to evaluate the value of door-to-door distribution of informational literature. *Resour Conserv Recycl*, *104*, 162–71. https://doi.org/10.1016/j.resconrec.2015.08.012

Xu, L., Ling, M., & Wu, Y. (2018, 07). Economic incentive and social influence to overcome household waste separation dilemma: A field intervention study. *Waste Manag*, *77*, 522–31. https://doi.org/10.1016/j.wasman.2018.04.048

Xu, L., Yang, H., & Ling, M. (2023, May). Factors moderating the effect of financial incentives on household recycling: A motivation crowding perspective. *Environ Impact Assess Rev*, *100*, 107078107078. https://doi.org/10.1016/j.eiar.2023.107078

Zhang, X. (2023, Dec). A systematic literature review on individuals' waste separation behavior. *Resources, Environment and Sustainability*, *14*, 100137100137. https://doi.org/10.1016/j.resenv.2023.100137

Zhang, Z., & Wang, X. (2020, Nov). Nudging to promote household waste source separation: Mechanisms and spillover effects. *Resour Conserv Recycl*, *162*, 105054105054. https://doi.org/10.1016/j.resconrec.2020.105054

Zhang, M., Ionkova, K., & Boukerche, S. (2025). *How the World Bank is tackling the growing global waste crisis [Internet]*. World Bank Blogs. World Bank Group. [cited 2025 Oct 8]. Available from: https://blogs.worldbank.org/en/sustainablecities/how-the-world-bank-is-tackling-the-growing-global-waste-crisis

Appendix A: A summary of the results

Study	N	Targeted or mentioned waste fraction(s)	RQ1: Intervention(s)	RQ2: Measurement approach	RQ2: Key quantitative findings	RQ2: Qualitative outcomes	RQ3: Barriers identified
Atthirawong (2016)	400	Solid waste	Communication campaign	ANOVA, Multiple Regression	Not quantified	Key factors: campaign perception and resident age	Lack of Infrastructure/Facilities
Bernstad et al. (2013)	630	Food waste, glass, paper, plastic, metal, newspapers, batteries, residual waste	Communication campaign	Weighing, composition analysis	Higher food waste separation; Lower impurities; Reduced residual waste	Door-stepping effects not durable; Demonstration more important than information	Space Constraints; System/Policy Issues
Bian et al. (2022)	5248	Food waste, residual waste	Infrastructure development	Pre-post tonnage measurements	No significant impact	Not provided	Practical Implementation Issues; Lack of Infrastructure/Facilities; Convenience/Time
Bortoleto and Hanaki (2007)	288	Solid waste (e.g. plastic, paper)	Communication campaign; Interactive/engagement activity; Infrastructure development; System-level reformation	Questionnaire	Not quantified	Changed citizen perceptions and practices	Practical Implementation Issues; Lack of Infrastructure/Facilities; Space Constraints
Bowman et al. (1998)	~1200	Not specified	Communication campaign; Social norm intervention	Behavior monitoring and surveys	Increased recycling in experimental groups	Better at improving existing recyclers than converting non-recyclers	Practical Implementation Issues; Lack of Infrastructure/Facilities; Convenience/Time; Knowledge/Awareness/Education
Bryce et al. (1997)	401	Solid waste	Communication campaign; Infrastructure development	Participation rate measurement	Verbal commitment: 6.31 weeks; Non-verbal: 5.59 weeks; Combined: 7.41 weeks	Verbal commitment effective; Financial commitment ineffective	Practical Implementation Issues; System/Policy Issues; Economic Factors
Chase et al. (2009)	258	Glass, plastic, paper, cardboard, batteries, cans, compost	Communication campaign	Campaign and follow-up surveys	University: plastic (77.6%), paper (76.2%); Community: aluminum (84.7%), plastic (78.4%)	Increased recycling-health awareness; Improved recycling efforts	Practical Implementation Issues; Lack of Infrastructure/Facilities; Convenience/Time; Knowledge/Awareness/Education
Chen et al. (2022)	1916	Not specified	Communication campaign	Pre-test/post-test	Significant improvement in waste separation habits	Celebrity modeling more effective than education; Different effects based on concern level	Attitudes/Motivation
Cole et al. (2014)	Not specified	Paper, cardboard, glass, metals, mixed plastic items, organic waste, mixed waste	Infrastructure development; System-level reformation	Monthly waste changes	Intervention 1: + 91t/month recycling, −120t/month residual; Intervention 2: + 111t/month recycling, −176t/month residual	Significant positive impact on recycling; Negative impact on residual waste	Not listed
Cotterill et al. (2009)	658	Glass bottles, cans, thin card, directories, textiles, paper, plastic bottles, batteries	Communication campaign; Infrastructure development	Participation rates over time	5.4% recycling improvement vs. control	Participation declined after three months	Practical Implementation Issues; Lack of Infrastructure/Facilities

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De Fano et al. (2022)	467	Plastics	Communication campaign	Questionnaire, multi-group analysis	Not quantified	Key influences: attitudes, peers, cost, facility availability; Social media ineffective	Not listed
De Gisi et al. (2017)	203	Solid waste	Communication campaign; Infrastructure development	Pre-post questionnaire	Not quantified	Information transfer ineffective; Technology viewed positively; Student awareness improved	Knowledge/Awareness/ Education; System/Policy Issues
Del Cimmuto et al. (2014)	183	Organic matter, paper, glass, plastic, metal, residues	Communication campaign; Infrastructure development	Questionnaire	Correct implementation: 87.5% (intervention) vs. 63% (control), $p < 0.001$	Door-to-door collection preferred; Higher information satisfaction (84% vs. 21%)	Lack of Infrastructure/Facilities; Convenience/Time; Knowledge/Awareness/ Education
Dhokhikah et al. (2015)	300	Food waste, plastic waste, paper waste, glass, woods (coconut shell and bamboo packaging)	Communication campaign; Infrastructure development	Questionnaire, regression	Not quantified	Key factors: age, knowledge, environmental cadre, waste bank availability	Practical Implementation Issues; Lack of Infrastructure/ Facilities; Convenience/Time; Knowledge/Awareness/ Education; Space Constraints
Díaz et al. (2012)	5644	Hazardous waste, glass, paper, container, residual	Communication campaign	Questionnaire	80% separated paper, glass, and containers	Increased recycling center usage and hazardous waste knowledge	Convenience/Time; Knowledge/Awareness/ Education; Space Constraints; Attitudes/Motivation
Dutta-Powell et al. (2023)	600+	Dry waste (e.g. cardboard, paper, plastic bottles, old clothes), wet waste, hazardous waste	Communication campaign; Infrastructure development; Performance-based messaging	Three-week waste collection	Improved dry waste segregation; No wet waste effect; Reduced hazardous waste errors	Both treatments effective for dry and hazardous waste	Practical Implementation Issues; Lack of Infrastructure/ Facilities; Space Constraints; System/Policy Issues
Flygansvær et al. (2021)	176	Food waste, plastic packaging, paper, cardboard, glass, metal, batteries, residual waste	Communication campaign; Infrastructure development	Waste analysis	Food waste sorting: + 17%; Glass/ metal in residual: −29%; Contaminated bags: −10%	Improved recycling behavior and sorting quality	Not listed
Geislar (2017)	370	Organic waste	Communication campaign; Infrastructure development	Wilcoxon tests	Food scrap separation: 2.4 to 3.8 ($p < .001$, d = 0.95); Norm messaging: $p = .035$, d = 0.22	Infrastructure: large effect; Social interventions boosted participation	Lack of Infrastructure/Facilities
Hernández et al. (1999)	410	Organic, recyclable and unusable	Economic incentive; Infrastructure development; System-level reformation	Qualitative research and survey	50% practiced waste separation	Key predictors: guideline knowledge, collection satisfaction, agreement with municipal collection	Practical Implementation Issues; Convenience/Time; System/Policy Issues; Economic Factors; Social/ Cultural Factors

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Hotta and Aoki-Suzuki (2014)	Not specified	Combustible waste, incombustibles, hazardous waste, cans, bottles, PET bottles, metal, paper, containers, packaging, textile, bulky waste	Communication campaign; Economic incentive	Interviews and waste statistics	Yokohama: 42% reduction; Kamakura: Initial success then failure	Yokohama closed incinerators; Kamakura restarted closed facility	Practical Implementation Issues; Lack of Infrastructure/ Facilities; System/Policy Issues; Economic Factors; Social/Cultural Factors
Huang et al. (2018)	~2300	Food waste	Communication campaign	Waste quantity and quality measures	Leaflets: no impact; Neutral interactions: small impact; Positive interactions: significant impact	Positive human interaction important factor	Not listed
Karout and Altuwaijri (2012)	306	Paper, metals, glass, plastic, batteries	Communication campaign	Pre-post questionnaire	Knowledge improved ($p < 0.005$); Hazardous waste awareness: 0.6% to 91.4%	Significant knowledge improvement across all items	Knowledge/Awareness/ Education
Kattoua et al. (2019)	246	Paper, cardboard	Communication campaign	Questionnaire	96% not participating in recycling	Not provided	Practical Implementation Issues; Lack of Infrastructure/ Facilities; Knowledge/ Awareness/Education; Space Constraints; System/Policy Issues
Komatsu et al. (2022)	1236	Plastic	Communication campaign	Survey	Text-only: negative effects; Text + illustrations: positive effects	Visual presentations more effective than text alone	Practical Implementation Issues; Convenience/Time; Attitudes/Motivation; Social/ Cultural Factors
Kutralam-Muniasamy et al. (2023)	2110–4368	E-waste, paper, cardboard, glass bottles, aluminum cans, plastics, packaging, tetrapack containers, batteries	Infrastructure development; Economic incentive; Reverse logistics/product return; Communication campaign	Yield assessment, trend analysis	263.63t waste diverted; 304.02t CO2-eq reduction; 65% growth	E-waste most abundant (63%); Packaging waste least abundant	Not listed
Lakhan (2016)	186	Paper, paper packaging, plastics, metals, glass, hazardous waste, organics	Infrastructure development; Communication campaign	Waste audits and surveys	Greatest increase with bins at building entrance	Convenience alone insufficient; Paper packaging most improved; No sustained increase	Practical Implementation Issues; Convenience/Time
Li (2003)	391	Household waste	Economic incentive	Logistic regression, chi-square	Low-income households more active in low-value recycling	Cash incentives effective, especially for lower incomes	Attitudes/Motivation; Economic Factors; Social/ Cultural Factors
Li et al. (2021)	13,282	Food waste	Incentive condition removal	Tonnage and interviews	20–25% reduction without incentives; 50–55% reduction in normal-hour participation	25–30% continued off-hours; 10–35% increase with staff return	Practical Implementation Issues; Convenience/Time; Attitudes/Motivation;

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							Economic Factors; Social/ Cultural Factors
Liang et al. (2024)	675	Kitchen waste, valuable waste, hazardous waste, other waste	Infrastructure development; Communication campaign	Questionnaire	TFS, CSG, UCS reduced low-level and increased high-level behaviors; 75% mediating effect	Waste separation habit key mediator; VDS nudge ineffective	Not listed
Linder et al. (2018)	474	Food waste, household waste	Communication campaign	Difference-in-difference analysis	Increased food waste collection in treatment group	Effects lasted 8 + months	Practical Implementation Issues; Lack of Infrastructure/ Facilities; Convenience/Time; Knowledge/Awareness/ Education
Ling et al. (2023)	7195	Dry waste	Communication campaign; Economic incentive	Waste weighing, survey	Normative feedback: 40% effect initially, 20% later	Normative feedback most effective; Effects decreased over time	Not listed
Liu et al. (2023)	560	Solid waste	Infrastructure development; System-level reformation	Questionnaire	96.43% participation vs. 26.4% seven years earlier	Positive attitudes promoted successful sorting	Practical Implementation Issues; Convenience/Time; Knowledge/Awareness/ Education
Löytty et al. (2023)	3	Food waste	Interactive/engagement activity	Temperature monitoring, self-assessment	Not quantified	ICT system effective; Data useful for end-users but not vital for authorities	Practical Implementation Issues; Lack of Infrastructure/ Facilities; System/Policy Issues
Manomaivibool et al. (2018)	869	Organic waste	Infrastructure development; System-level reformation; Interactive/engagement activity; Economic incentive	Waste pile measurement	91.6% using composters; 96.7% separated recyclables; 48% separated 4 + types	Most sorted: plastic bottles (88%), glass (73%), plastic bags (55%)	Practical Implementation Issues; Space Constraints; System/Policy Issues
Meen-Chee and Narayanan (2006)	760	Paper, cardboard, plastic, metal, glass	Communication campaign	Survey with covariate analysis	73.8% higher recycling odds in campaign-aware participants	Environmentally conscious with storage space more likely to recycle; Gender/ distance not predictive	Practical Implementation Issues; Knowledge/ Awareness/Education; Space Constraints
Mehner et al. (2020)	93	Waste to incinerate, plastic packaging, paper packaging, newsprint, metal packaging, glass packaging	Communication campaign; Infrastructure development	Pick analyzes and surveys	Mis-sorted items: 38.6% to 36.3% (not significant); Metal packaging: 40% reduction	Improved sorting self-assessment; Increased knowledge and behavior	Lack of Infrastructure/Facilities; Convenience/Time; Space Constraints
Nepal et al. (2023)	993	Solid waste	Communication campaign; Infrastructure development	Three-round survey (control/ intervention)	7.6–11.4% increase in waste collection; 12.4–21.1% cleanliness improvement	Awareness insufficient for source segregation changes	Practical Implementation Issues; Lack of Infrastructure/ Facilities; Attitudes/ Motivation; System/Policy Issues; Economic Factors

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Nguyen (2022)	192	Compostable, non-compostable and recyclable waste	Infrastructure development; System-level reformation	Questionnaire	41.5% continued sorting years after intervention	Not provided	Social/Cultural Factors
Nowakowski et al. (2021)	251	E-waste	Infrastructure development	Surveys and field study	87% aware of requirements; 61% noticed containers	Containers visible but broader campaigns lacking; 70% dispose e-waste ≤ 1/year	Practical Implementation Issues; Space Constraints; System/Policy Issues
Okonta and Mohlalifi (2020)	480	Recyclable (e.g. plastic, paper, glass, aluminum), other waste	Economic incentive; Infrastructure development; Communication campaign; System-level reformation	Questionnaire	Recycling rated lower than water/electricity savings	Communication preferences varied by education; Key drivers: education, finances, facilities	Lack of Infrastructure/Facilities
Omran et al. (2009)	389	Solid waste	Communication campaign	Postal questionnaire	91% campaign awareness; 59.9% non-participation	Education and accessibility key for positive attitudes	Practical Implementation Issues; Lack of Infrastructure/Facilities; Convenience/Time; Knowledge/Awareness/Education; Space Constraints; Attitudes/Motivation; Economic Factors
Otto et al. (2018)	653	E-waste	Economic incentive; Interactive/engagement activity	Survey and e-waste collection	14-fold increase with structural facilitation; 2x increase with information/social enticement	Significant differences between structural interventions	Lack of Infrastructure/Facilities; Convenience/Time; Economic Factors
Park and Lah (2015)	Population	Paper, cans, bottles, metal, plastic, textile, other recyclable items	Economic incentive	Annual recycling rate	Increased from 6.3% to 47%	Not provided	Convenience/Time; Economic Factors
Pegels et al. (2022)	5857	Plastic, paper, glass, metal, organic	Communication campaign	Randomized controlled trial	50% reduction in mixed waste bags; Effects persisted 6 months	No added effect from empathetic messages or reminders	Practical Implementation Issues; Knowledge/Awareness/Education; Space Constraints; System/Policy Issues
Pongpunpurt et al. (2021)	484	Plastic	Infrastructure development	Questionnaire, waste weighing	Not quantified	Bin location critical for collection amount	Not listed
Pongpunpurt et al. (2022)	321	Organic waste, recyclable waste (e.g. paper, glass, metal, plastic, clothes), hazardous waste, other waste	Infrastructure development; Communication campaign	Questionnaire, post-hoc tests	Knowledge ($\beta = 0.653$, $p < 0.001$), norm ($\beta = 0.160$, $p < 0.05$) predicted intention; Intention influenced behavior ($\beta = 0.804$, $p < 0.001$)	House ownership significantly impacted sorting behavior	Not listed

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Read, (1999)	~7386	Solid waste (rubbish and recycled; paper, bottles mentioned)	Communication campaign	Pre-post tonnage comparison	Weekly recycling: 107 to 132 tonnes; Recycling rate: 13% to 17%	Not provided	Practical Implementation Issues; Convenience/Time; Knowledge/Awareness/ Education; Attitudes/ Motivation
Read et al. (2009)	5176	Dry recyclables, organic waste, other waste	Communication campaign	Monthly household waste averages	0.5 kg reduction/household/week (~2%); Control areas: 0.5 kg increase	Best activities: mail registrations, composting, doorstepping	Not listed
Remr (2018)	1579	Household waste	Communication campaign	Survey	69% wanted financial rewards; 63% intrinsically motivated	20% responsive only to direct incentives	Lack of Infrastructure/Facilities; Convenience/Time; Attitudes/Motivation
Roe et al. (2022)	40	Food waste	Interactive/engagement activity	Food waste data collection	78.8% reduction in plate waste; 20% decline in landfill/sewer waste	Reduced avoidable food waste; No effect on preparation/storage waste	Attitudes/Motivation
Rousta et al. (2015)	507	Paper packaging, plastic packaging, metal packaging, textiles, batteries, light-bulbs, small electronics, cooking oils, food waste, newsprints, glass packaging	Infrastructure development; Communication campaign	Pick analysis	Miss-sorted waste: 55% to 39%; Packaging/newsprint: 30% decrease	Shorter distance to drop-off (2 km to 50 m) key factor	Not listed
Sabbir et al. (2023)	334	E-waste	Reverse logistics/product return; Economic incentive	Structural equation modeling	Attitude, subjective norm, perceived control predicted intentions	Financial benefits enhanced participation; Knowledge/ opportunities increased interest	Not listed
Schmidt et al. (2022)	305	Organic, non-organic recyclables (plastic, paper, metals)	Communication campaign; Infrastructure development	Questionnaire, fly-counts	Flies: −64% intervention, +52% control	Varied community improvements; Low-cost collection improved household waste management	Other
Shan et al. (2020)	147	Metal, paper, plastic	Economic incentive; Interactive/engagement activity	Questionnaire and waste measurement	847.9kg collected; Paper: 92.5%; Metal: 3x higher with more points	Increased recycling (>40% vs. 20% national); Reduced contamination (<2% vs. 40% national)	Not listed
Siepelmeyer and Otterbring (2022)	66	Not specified	Social norm intervention; Performance-based messaging	Initial and follow-up surveys	Social identity: 5.24 vs. 4.76 ($p = 0.06$); Known neighbors: 9.00 vs. 4.17 ($p = 0.005$)	Environmental intervention increased social identity and capital	Not listed
Sintov et al. (2019)	284	Food waste	Infrastructure development; Communication campaign	Baseline, midpoint, follow-up surveys	Positive spillover to energy/water conservation	Composting affected other environmental behaviors	Not listed

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Timlett and Williams (2008)	4000–5000	Paper, card, cans, plastic bottles, glass, textiles	Communication campaign; Economic incentive; Performance-based messaging	Set-out and contamination monitoring	Incentives/feedback halved contamination; Feedback: £ 0.50/household, £ 3/behavior change	Doorstepping ineffective; No change in participation (already high)	Knowledge/Awareness/Education
Tong et al. (2018)	500	Household waste	Communication campaign; Economic incentive	Participation and sorting rates	Significant improvement with gift incentives	Social Norm key for initial recycling; Perceived Behavioral Control important for behavior	Lack of Infrastructure/Facilities; Convenience/Time
Vining & Ebreo, (1989)	434	Glass, newspaper, cardboard, office paper, aluminum, tin cans, motor oil	Communication campaign	Pre-post questionnaires	Increased recyclable knowledge; More materials recycled per household	Environmental motives strengthened; Economic motives remained low	Practical Implementation Issues; Lack of Infrastructure/Facilities; Convenience/Time; Knowledge/Awareness/Education; Space Constraints
Welfens et al. (2016)	Not specified	Mobile phones	Reverse logistics/product return; Communication campaign	Interviews, surveys, World Cafés	Not quantified	Economic incentives critical; Clear procedures important; Youth less engaged	Practical Implementation Issues; Lack of Infrastructure/Facilities; Convenience/Time; Knowledge/Awareness/Education; Space Constraints; Attitudes/Motivation; Economic Factors; Social/Cultural Factors
Willman (2015)	260	Garbage, recyclables	Communication campaign	Recycling participation measures	Door-to-door: 50% conversion to larger carts vs. 5% in control	Accelerated cart adoption but no increase in household participation	Practical Implementation Issues; Knowledge/Awareness/Education
Xu et al. (2018)	188	Recyclables (e.g. cardboard, metal, glass, plastic bottles) hazardous waste, other dry waste, wet waste	Communication campaign; Economic incentive	Pre-post surveys	Economic inducement more effective than social mobilization	Economic rewards more effective for lower incomes; Less effect on higher incomes	Not listed
Xu et al. (2023)	2802	Dry waste (e.g. paper, metals, plastics)	Economic incentive; Infrastructure development; Performance-based messaging; Communication campaign	Pre–post behavior/motivation surveys	36% reduction in program impact due to motivation crowding	Increased recycling but reduced intrinsic motivation; Worse effects for environmentally-identified people	Practical Implementation Issues; Attitudes/Motivation; Economic Factors; Social/Cultural Factors
Zhang and Wang (2020)	11,193	Household waste	Communication campaign	Household waste sorting survey	0.243 points higher sorting frequency in pilot cities (3-point scale)	Young, female, educated, non-rural respondents more likely to separate waste	Lack of Infrastructure/Facilities; Attitudes/Motivation; Social/Cultural Factors